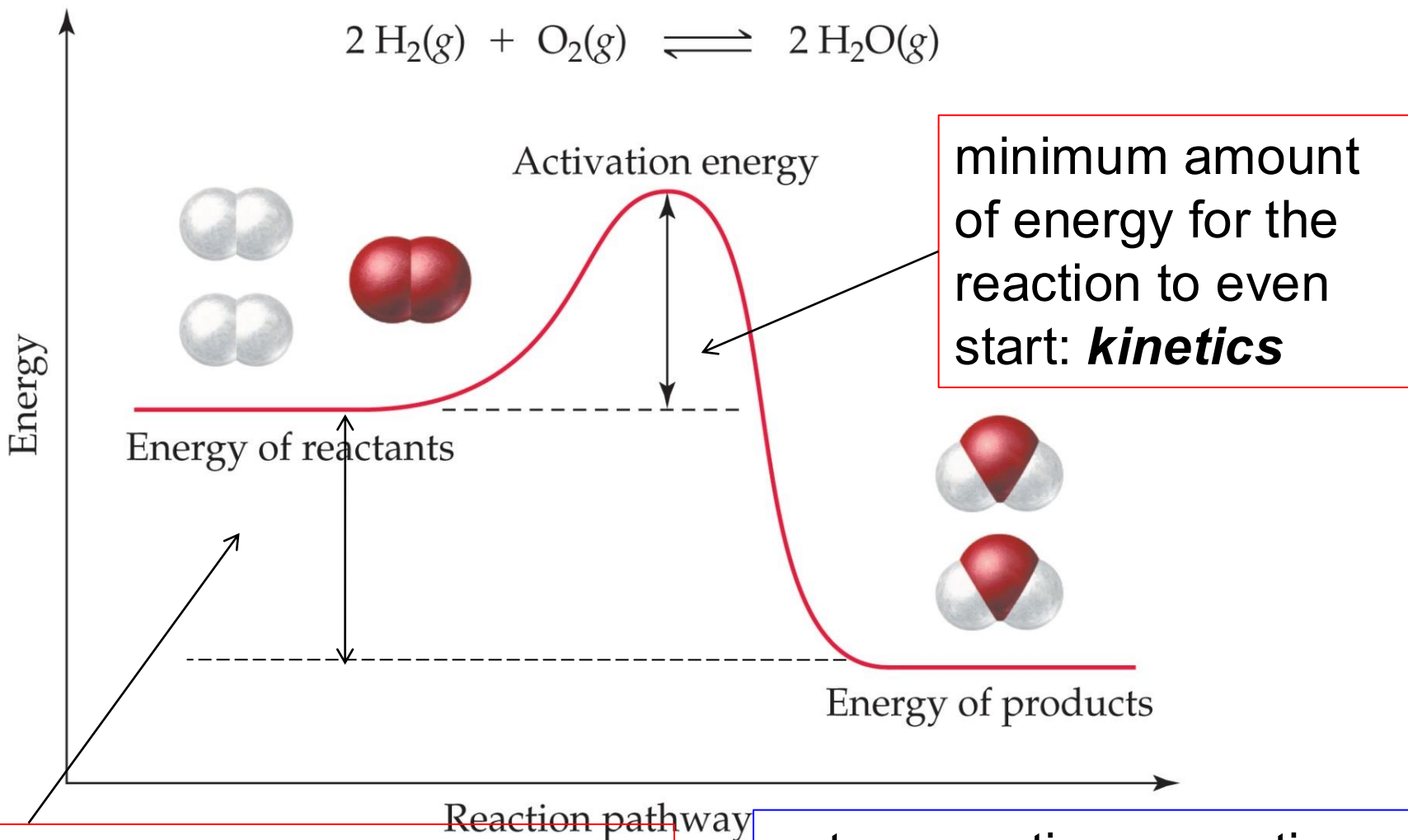


Why are Reactions Important?

Everything
is made by
chemical reactions

Understanding basic chemical reactions allow us to make new materials and **reproducibly** make known substances

How do reactions happen?

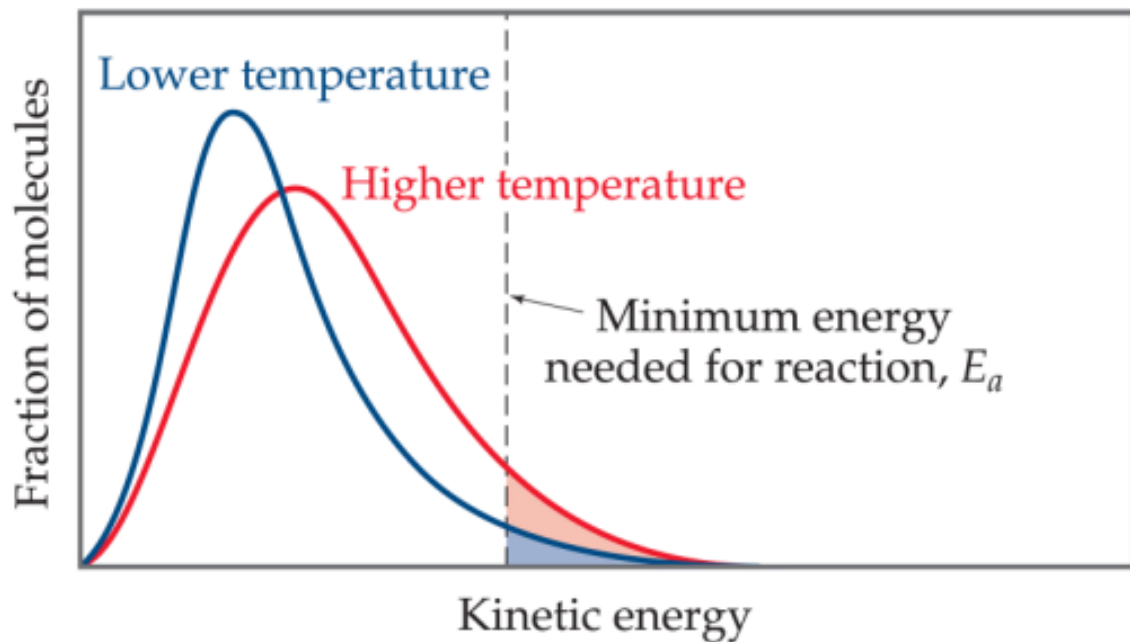


energy released upon reaction:
thermodynamics

How do reactions happen?

Breaking bonds **always** takes energy

1) Molecules are always in motion



The key is Energy

2) Add energy – heat, light for example

What is a Chemical Reaction?

- In a chemical reaction, one or more substances are converted into different substances.
 - The starting substance is a reactant.
 - The resulting substance is a product.
- We represent chemical reaction using a Chemical Equation as shown below.



- ❖ *In writing chemical reaction, it is important to show the physical states of each reactants and products.*
- ❖ *Of course the chemical reaction has to be completely balanced (number of atoms on either side the equation should be equal) – **conservation of matter**.*

Symbols Used in Chemical Equations

- Symbols used to indicate state after chemical.
 - (g) = gas; (l) = liquid; (s) = solid.
 - (aq) = aqueous = dissolved in water.
- Energy symbols used above the arrow for decomposition reactions.
 - Δ = heat.
 - $h\nu$ = light.
 - shock = mechanical.
 - elec = electrical.

- In a chemical reaction, any of the following process could take place.
 - Atoms of different elements can combine to make new compounds.
 - Molecules can combine to make bigger molecules.
 - Molecules can decompose into smaller molecules or atoms.
 - Atoms can be exchanged between molecules or transferred to another molecule.
 - Atoms can gain or lose electrons, turning them into ions.
 - Or changing the charge on ions that are already there.

But, **remember this**.... Elements are not ***transmuted*** during a reaction. (*in other words, nothing happens to the nucleus in a chemical reaction!*).

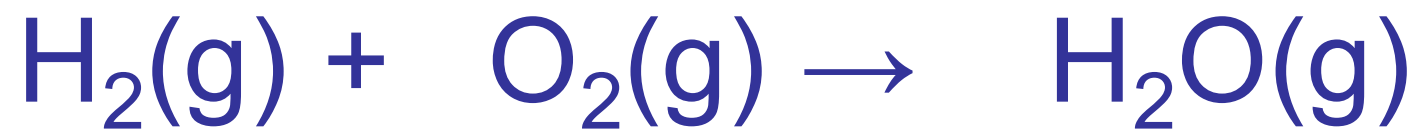
Balancing Chemical Equations

1. Count the number of atoms of each element on each side of the equation. Determine which need to be balanced.
 - Polyatomic ions may often be counted as if they are one “element”. Example, CO_3^{2-} , SO_4^{2-} , NO_3^- etc
2. Pick an element to balance.
 - Usually start with an atom in the most complicated molecule.
 - If an element is found in only one compound on both sides, balance it first.
 - Metals before nonmetals.
 - Leave elements (like O_2 and H_2) until last.
3. Put coefficients in front of the formula to bring the elements into balance. ***Never change a subscript in a compound.***

4. Move on to next element and balance. Repeat as necessary.
5. Make sure that all coefficients are whole numbers. If not, multiply all coefficients by the same number on both sides of the equation so that all the coefficients will be a whole number.

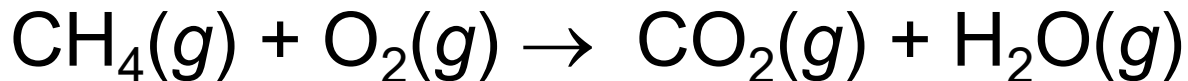
Fractions are acceptable, but not recommended: you can write $\frac{1}{2}$, but not 0.5 in front of a species.

You can master this only if you practice.



Example: Combustion of Methane

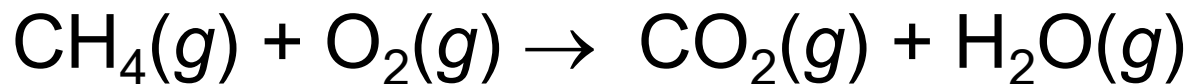
- Methane gas burns to produce carbon dioxide gas and gaseous water.
 - Whenever something burns it combines with $O_2(g)$.



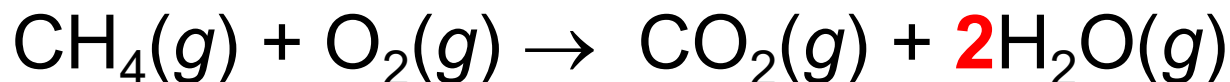
This equation reads “1 molecule of CH_4 gas combines with 1 molecule of O_2 gas to make 1 molecule of CO_2 gas and 1 molecule of H_2O gas”. The chemical equation is **NOT** balanced as you can see.

On the left side (Reactant side): 1 C, 4 H and 2 O

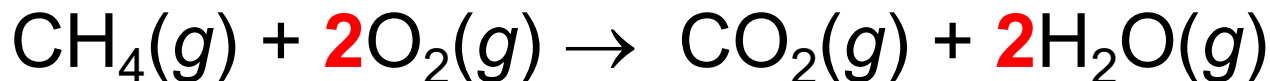
On the right side (Product side): 1 C, 2 H and 3 O



- The carbon atoms are already balanced. Oxygen exists in two compounds, so skip for now. Try to balance H.
- Put 2 as coefficient for water on the right.



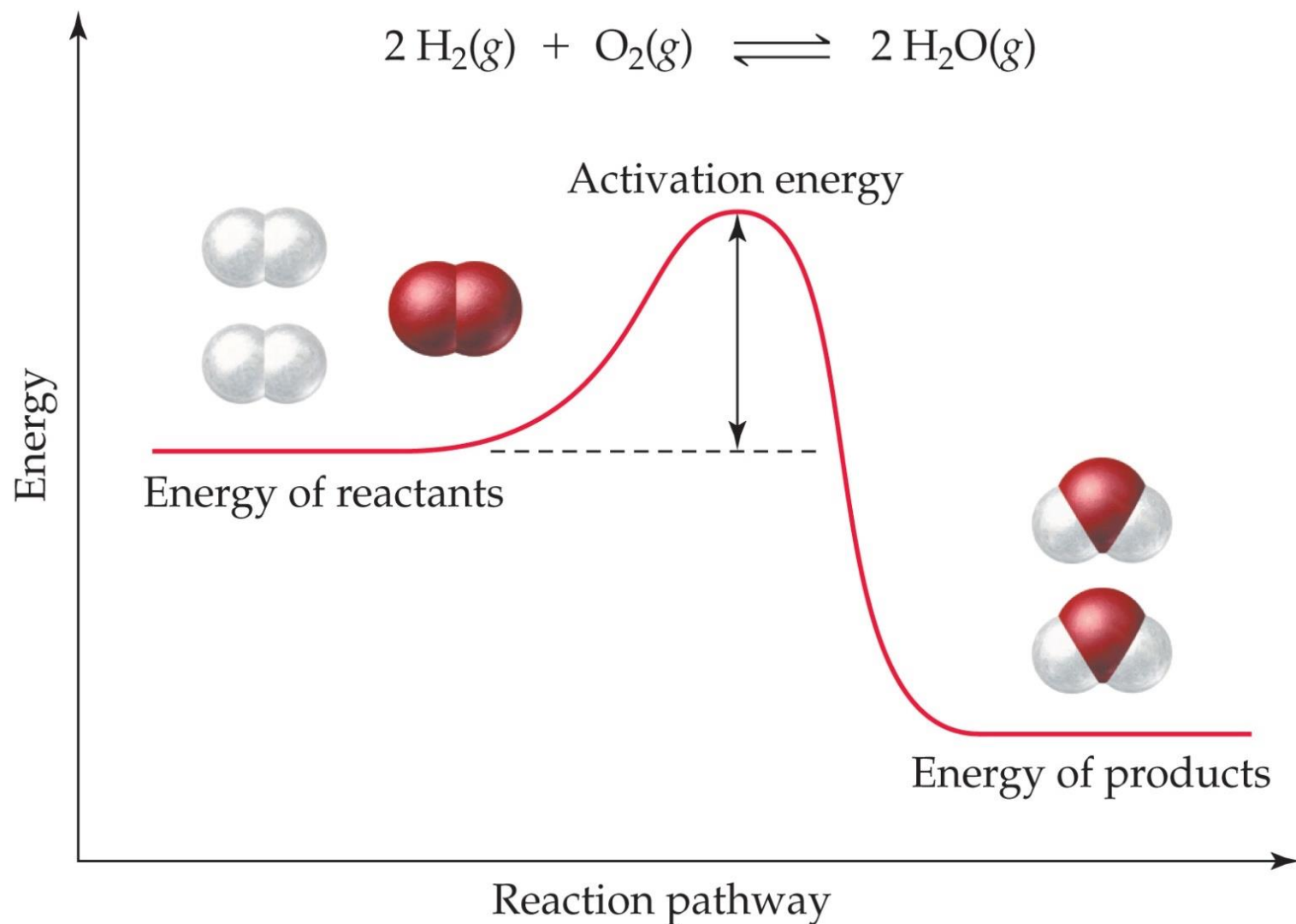
- Now, C and H are balanced, so focus on oxygen. To make oxygen even on both sides, put 2 in-front of $\text{O}_2(g)$ on the reactant side.



- Recheck:
 - On the left (reactant side): 1 C, 4 H and 4 O
 - On the right (product side): 1 C, 4 H and 4 O

How do reactions happen?

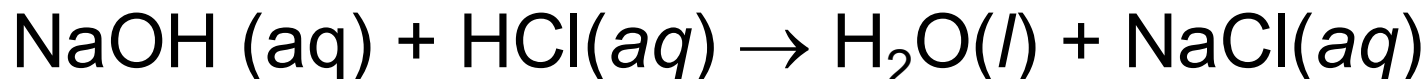
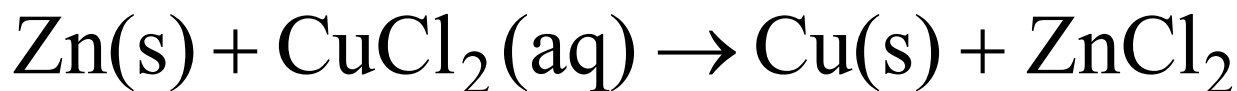
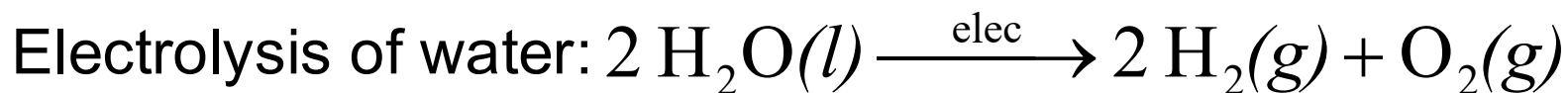
Thermochemistry



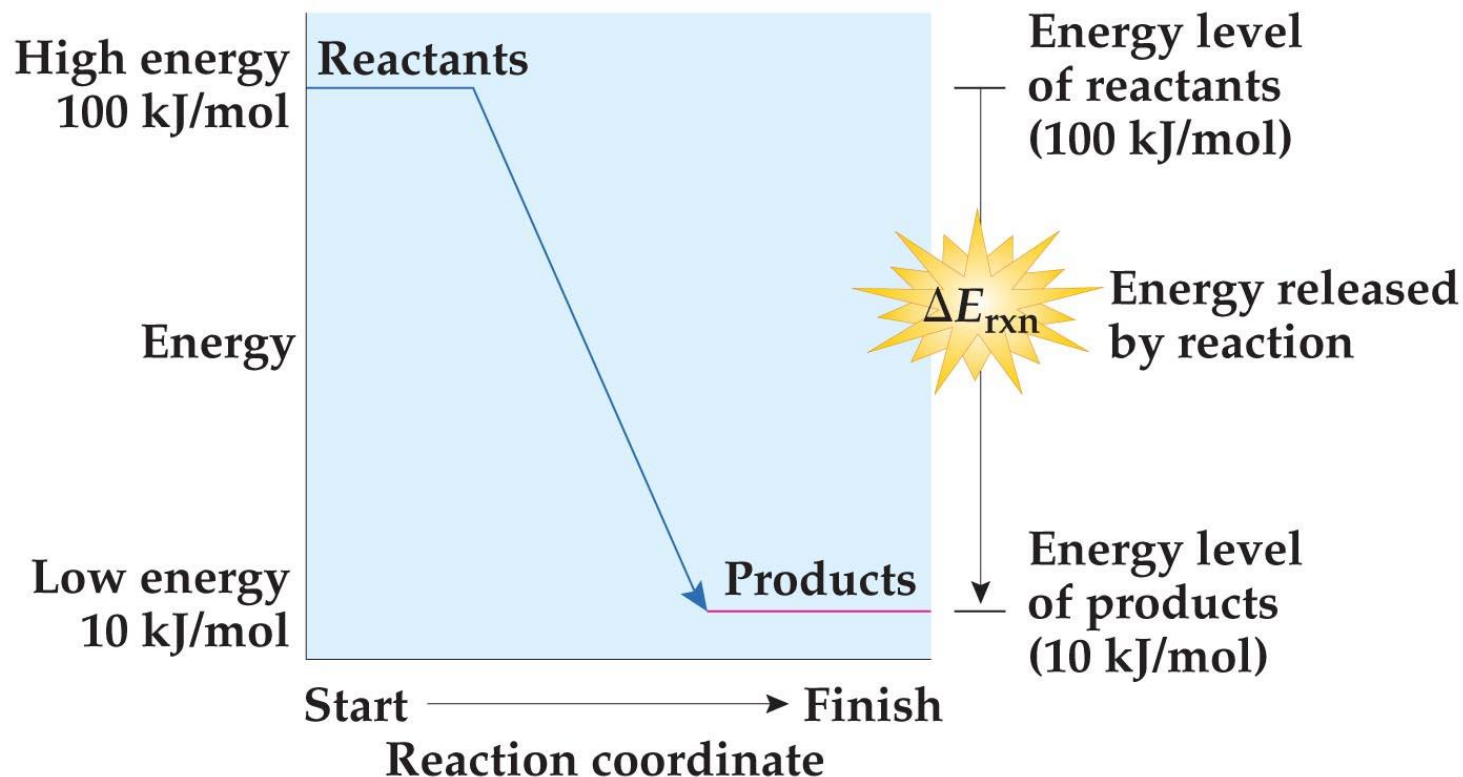
Remember: chemical reactions involve:

Reactant bonds breaking – requires energy

Product bonds forming – releases energy



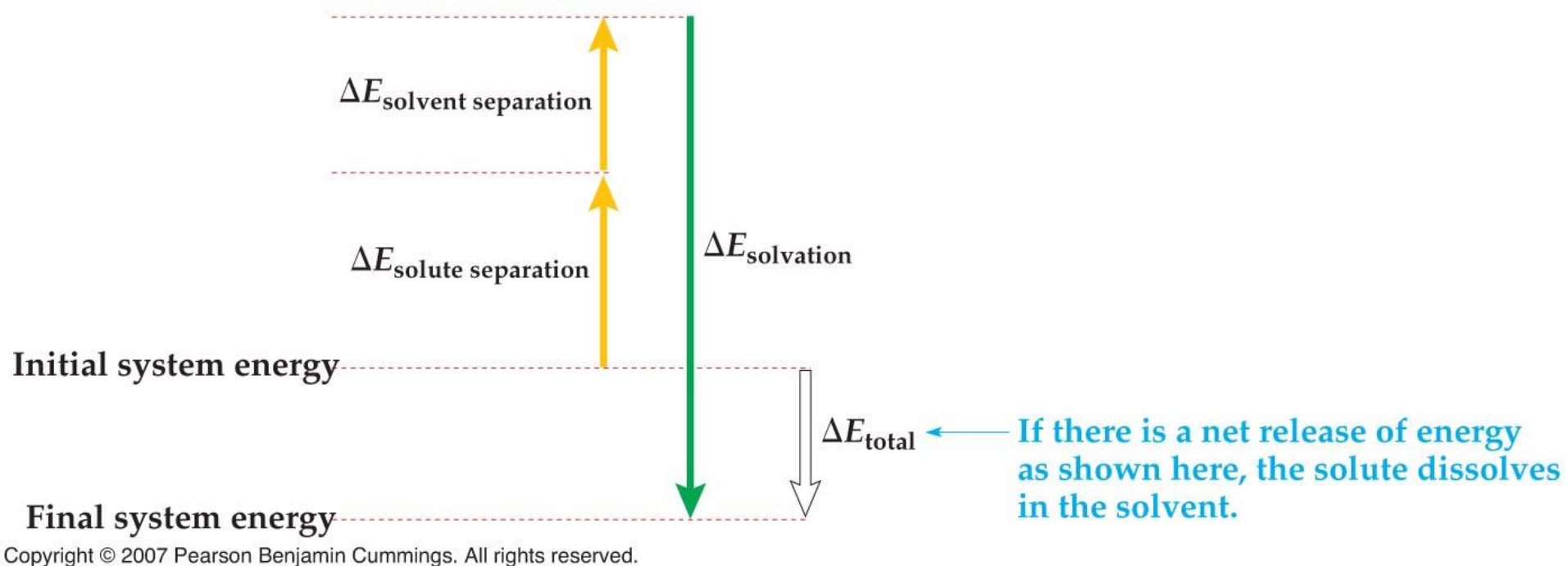
Energetics of reactions: 1



Reactions which release energy are called **exothermic** reactions.

$$\Delta E = E_{\text{products}} - E_{\text{reactants}} < 0$$

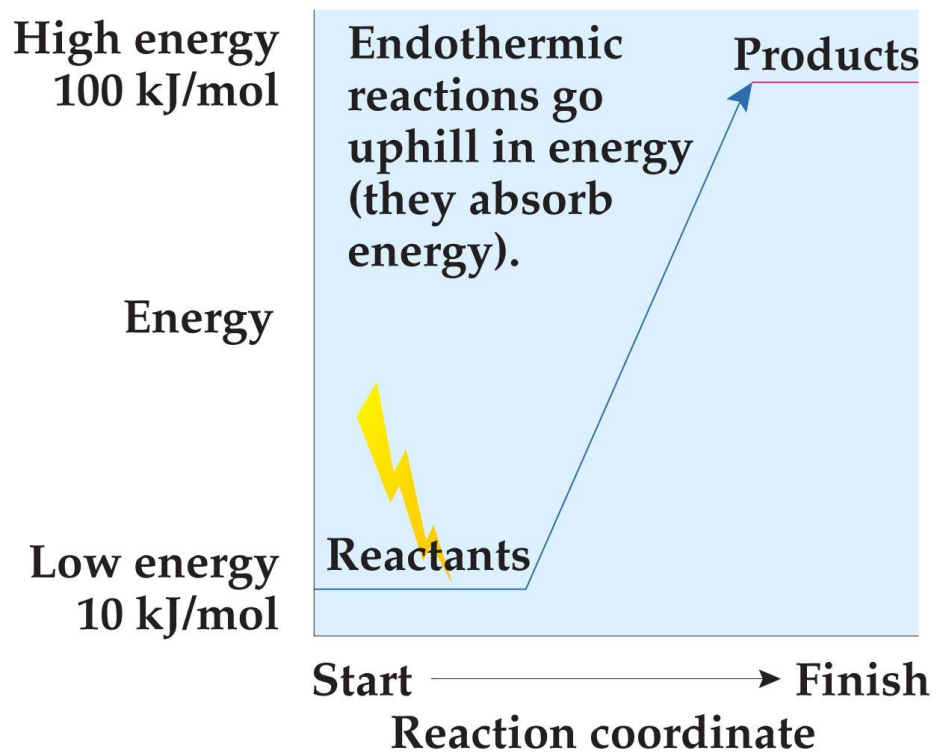
Will it dissolve?



If $\Delta E_{\text{total}} < 0$ (is negative), the substance will dissolve and the solution forms, the reaction is **exothermic**

Solution will also get warmer

Energetics of reactions: 2

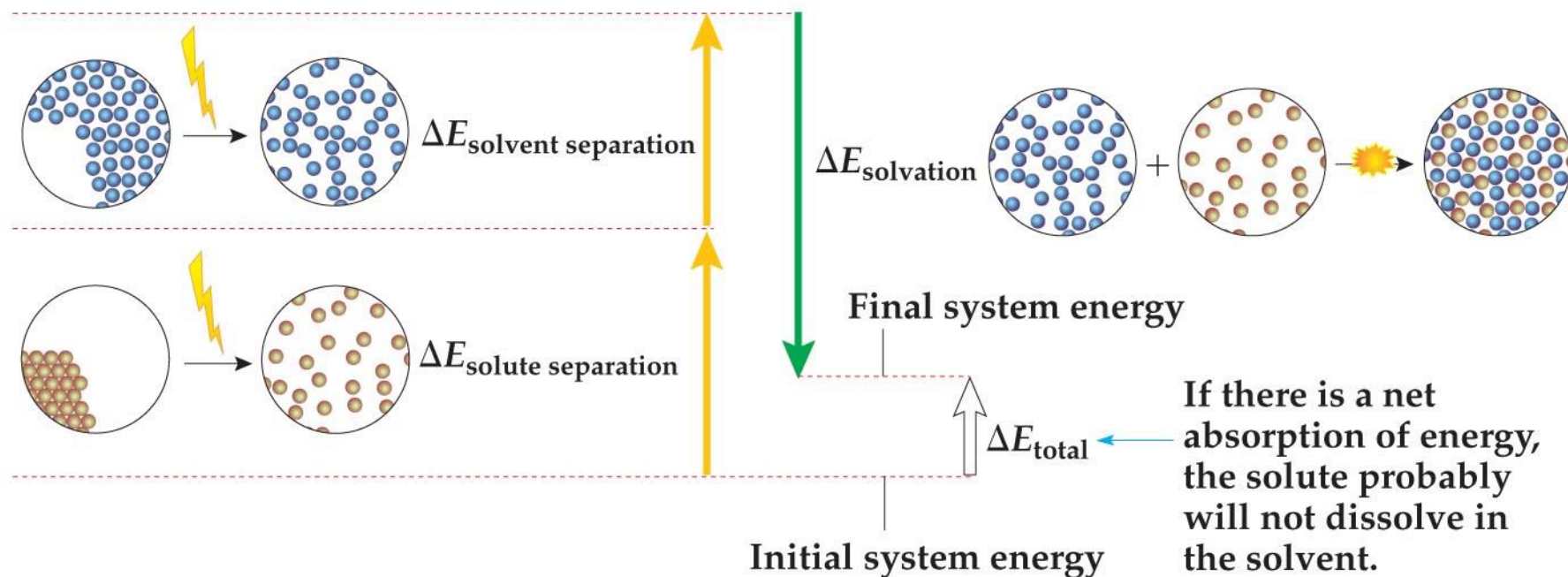


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Reactions which absorb energy are called **endothermic** reactions.

$$\Delta E = E_{\text{products}} - E_{\text{reactants}} > 0$$

Will it dissolve?



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If $\Delta E_{\text{total}} > 0$ (is positive), the substance will *probably* not dissolve, the reaction is **endothermic**

Solution will also get colder if the substance dissolves

Energetics of reactions: 3

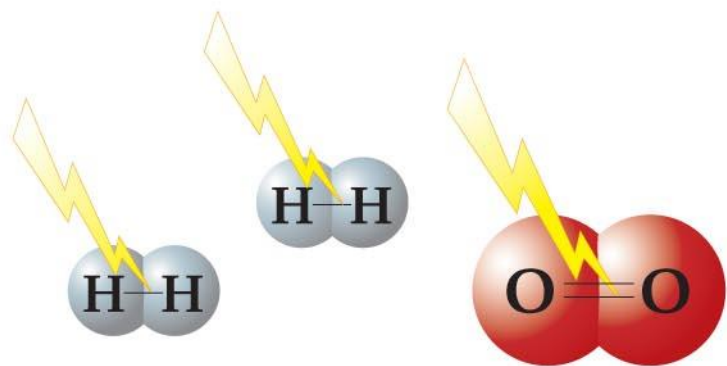
Reactions where the reactants and products are at the same energy are **energy neutral** or **thermal neutral** reactions.

Very rare!!!

$$\Delta E = E_{\text{products}} - E_{\text{reactants}} = 0$$

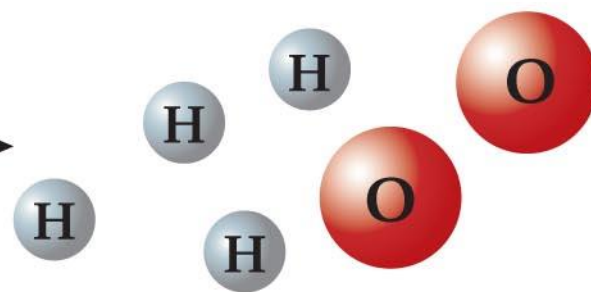
Another “hypothetical” reaction

Energy absorbed



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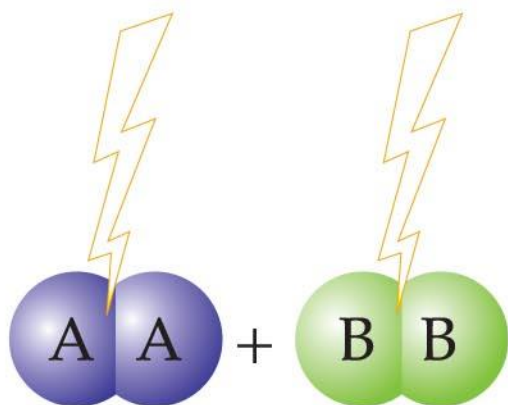
Chemical bonds broken



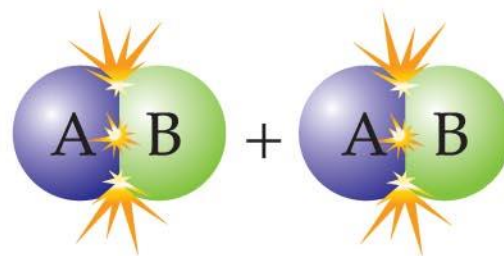
Chemical bonds formed,
energy released



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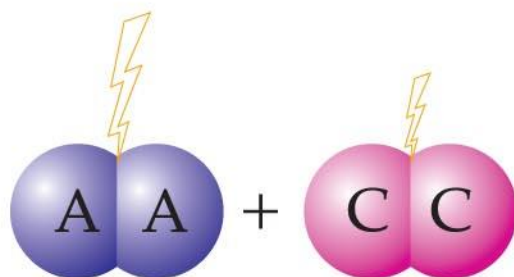


**Bond breaking absorbs
lots of energy.**

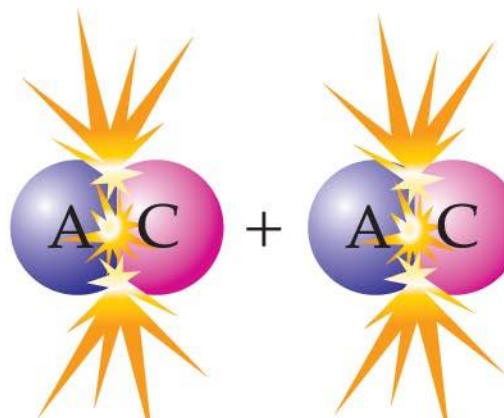


**Bond formation releases
little energy.**

} **Endothermic
reaction**



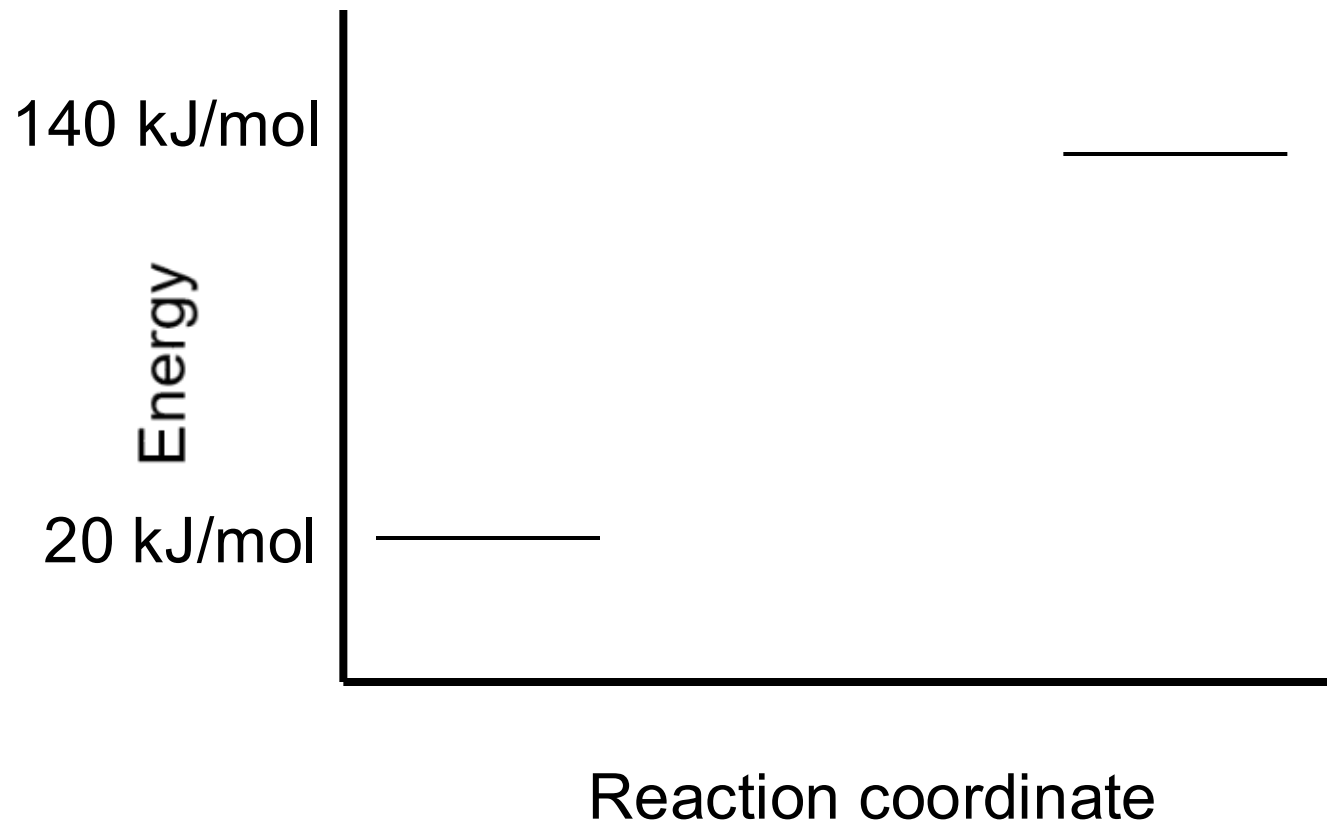
**Bond breaking absorbs
little energy.**



**Bond formation releases
lots of energy.**

} **Exothermic
reaction**

Another example



$$\Delta E_{\text{forward}} = -\Delta E_{\text{reverse}}$$

Is ice melting endothermic or exothermic?

Is water boiling endothermic or exothermic?

**All of this still doesn't tell us
how fast a reaction happens!!!**

Exothermic reactions can be fast!

Exothermic reactions can be slow!

Endothermic reactions can be fast!

Endothermic reactions can be slow!

Chemical kinetics

Why do some reactions happen quickly?

Why do some reactions happen slowly?

Chemical kinetics is the study of chemical reaction rates and how they change.

Macroscopic factors that affect the rate of a reaction

1. Physical state – gases tend to react faster than liquids
2. Concentration of reactants
3. Temperature
4. Catalyst

Multiple reaction steps

A reaction rarely involves only one step

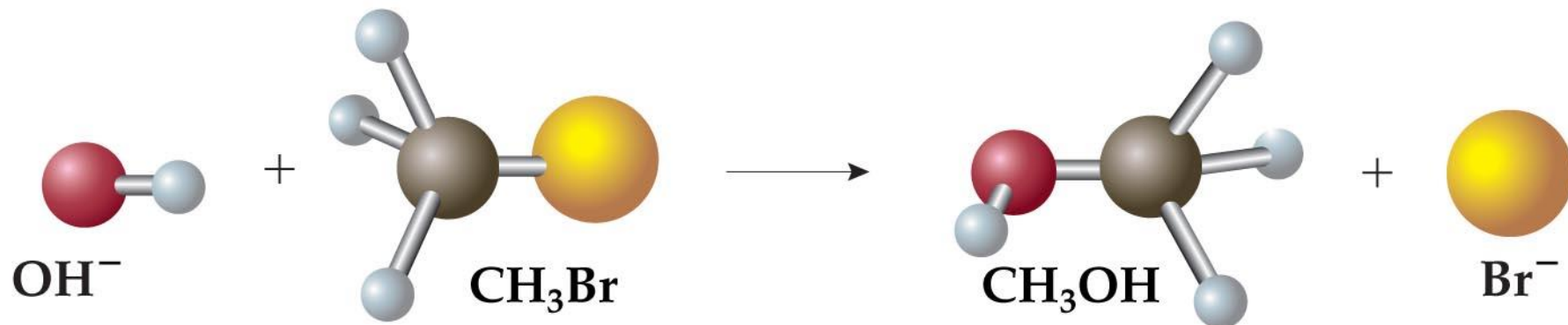
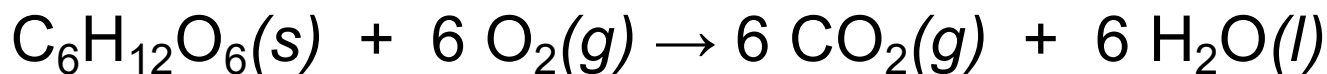
student in dorm room → student in lecture hall

This process is not a single step,
but is the net overall change from a series of steps:

student in dorm room → student at 1st bus stop
student at 1st bus stop → student at 2nd bus stop
student at 2nd bus stop → student in lecture hall

Chemical kinetics

A **reaction mechanism** is the detailed series of reactions steps that represents how reactants move from reactants to products.

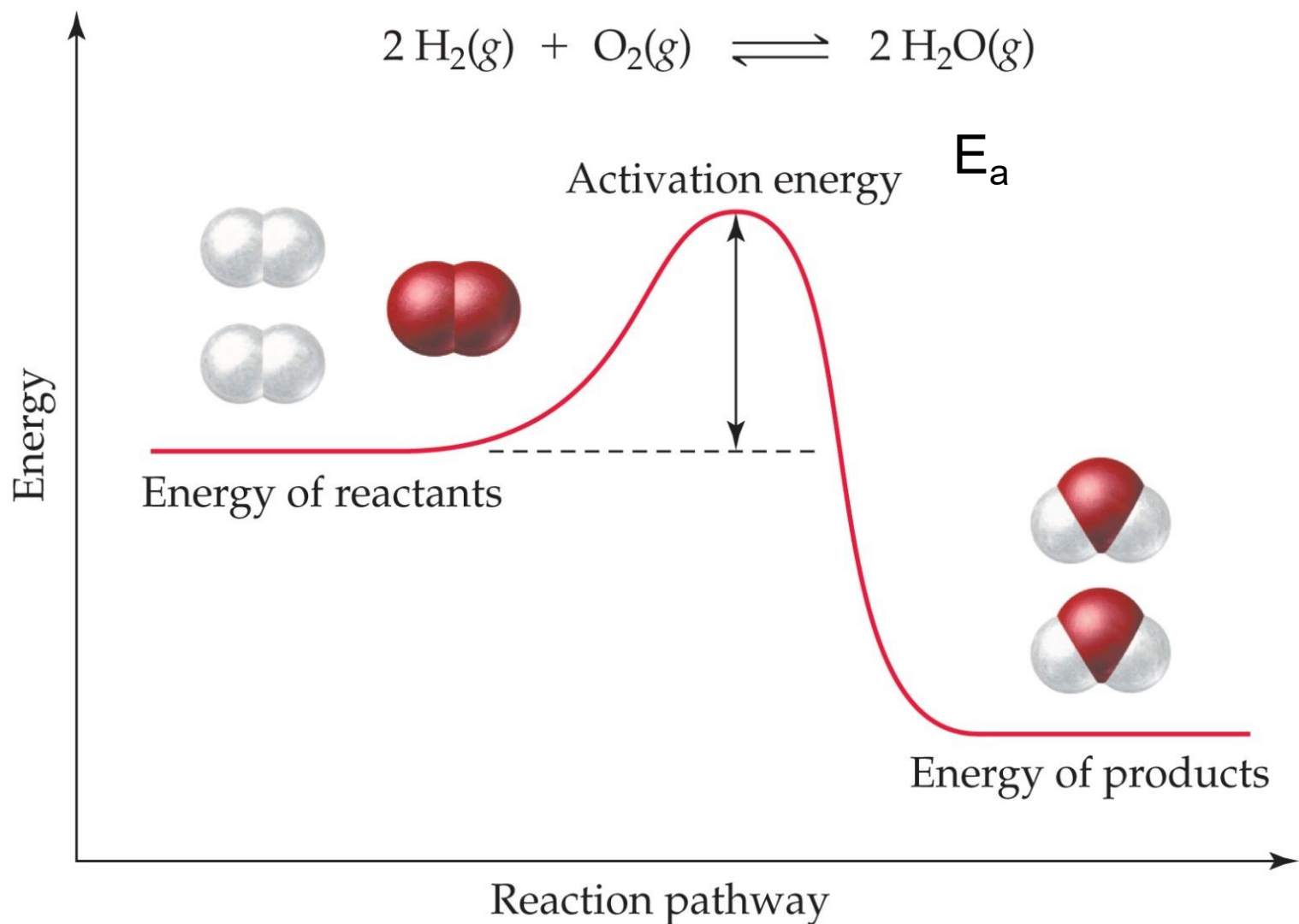


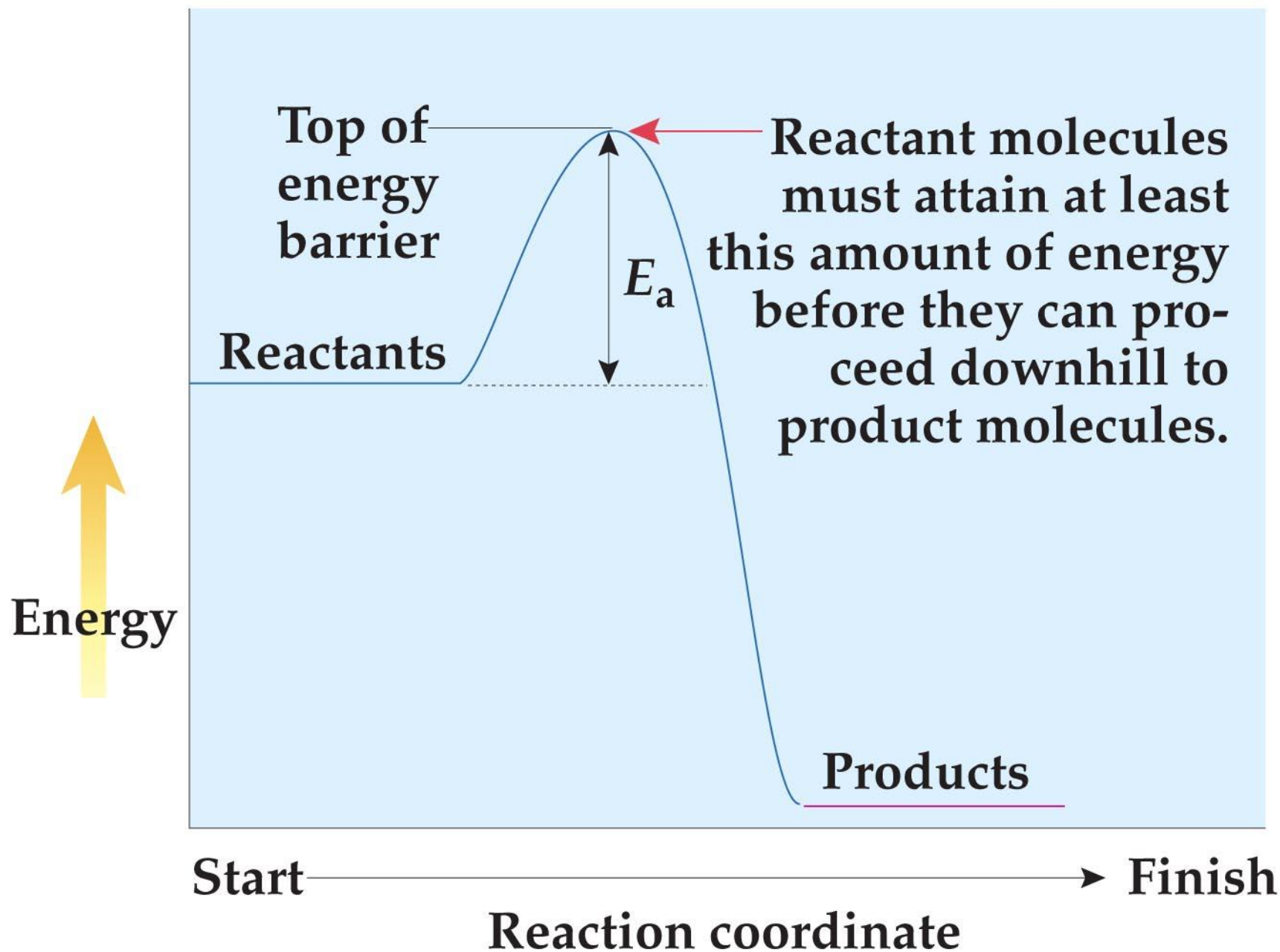
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Substitution reaction

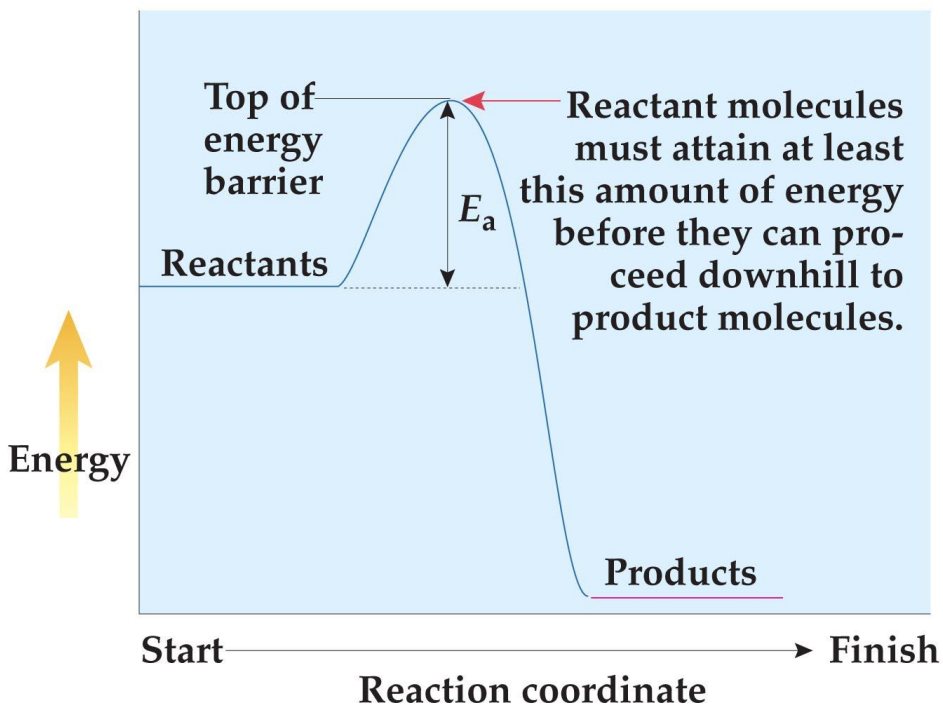
How do we figure out the reaction mechanism?

What determines the rate of a reaction (microscopic)?

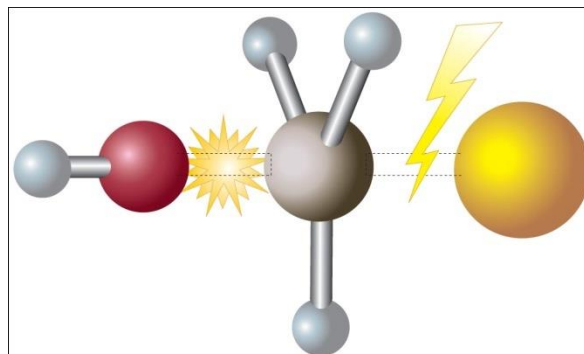




Transition state



The **transition state** is the high energy, unstable state of the molecules at the top of the barrier.

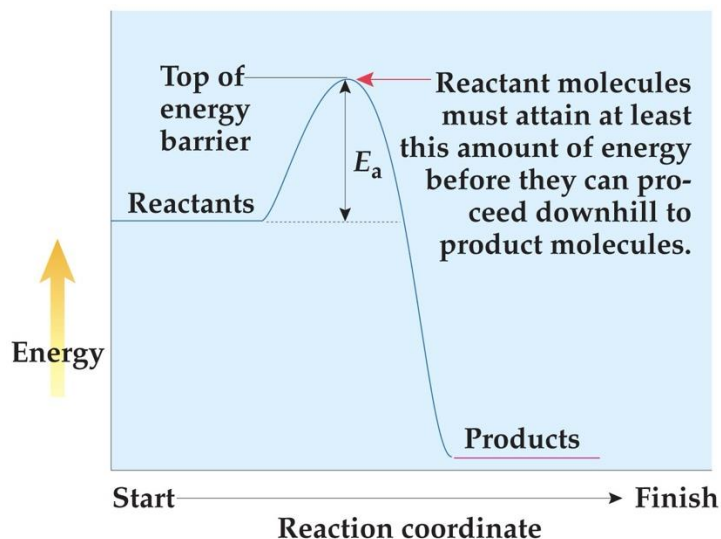


Reaction rate depends on E_a

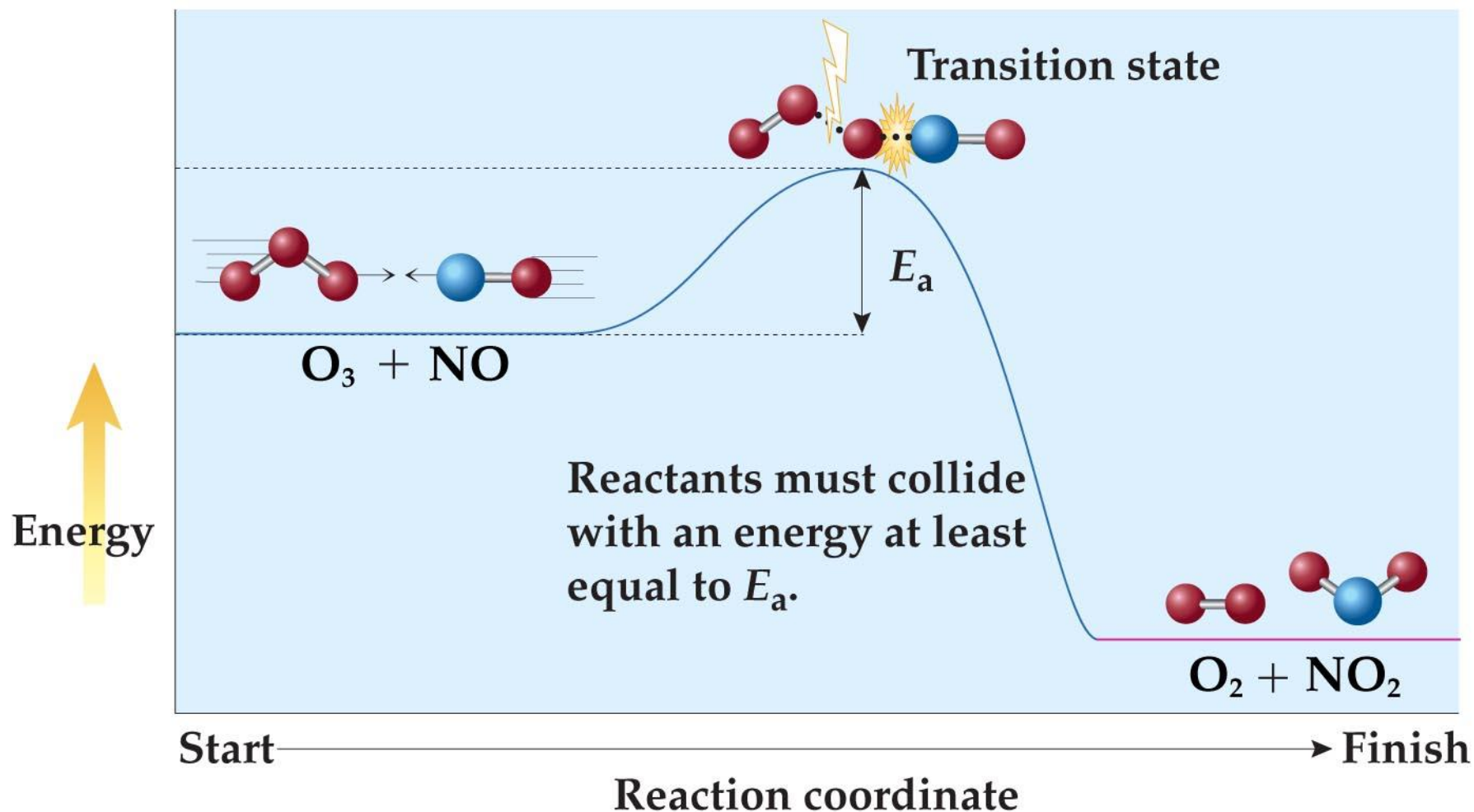
The higher the activation barrier,
the slower the reaction!

Large $E_a \rightarrow$ slower reaction

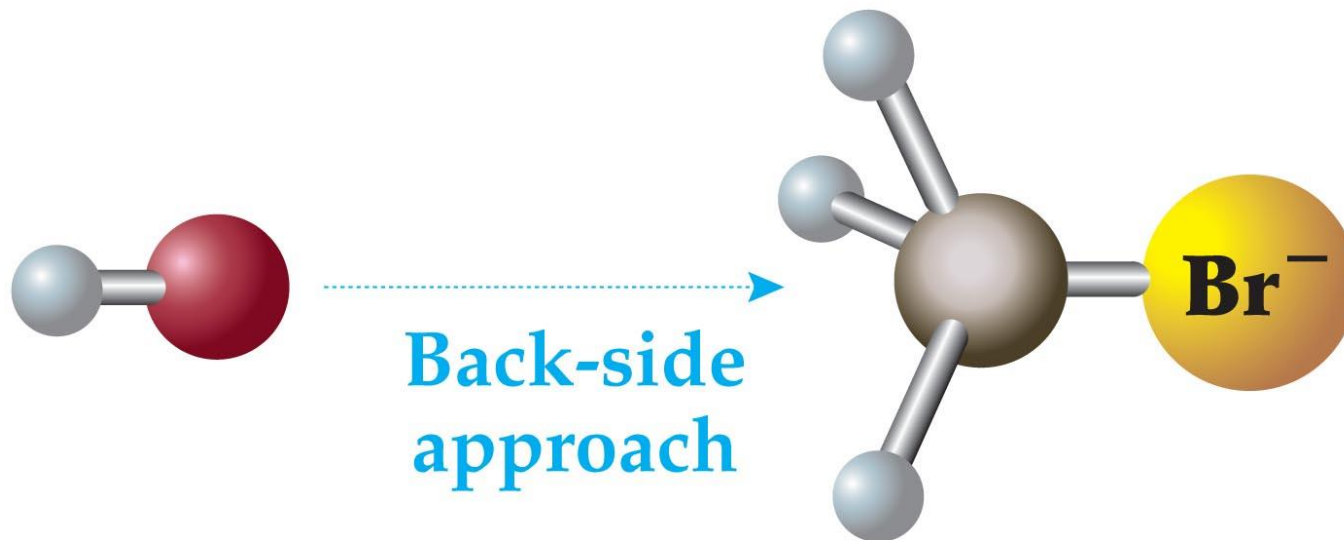
Small $E_a \rightarrow$ faster reaction



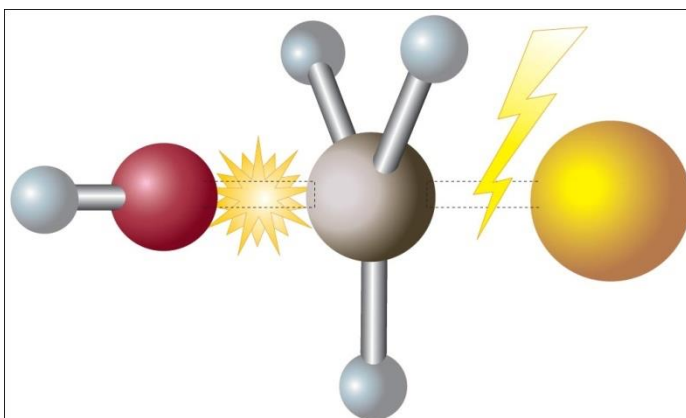
Orientation matters



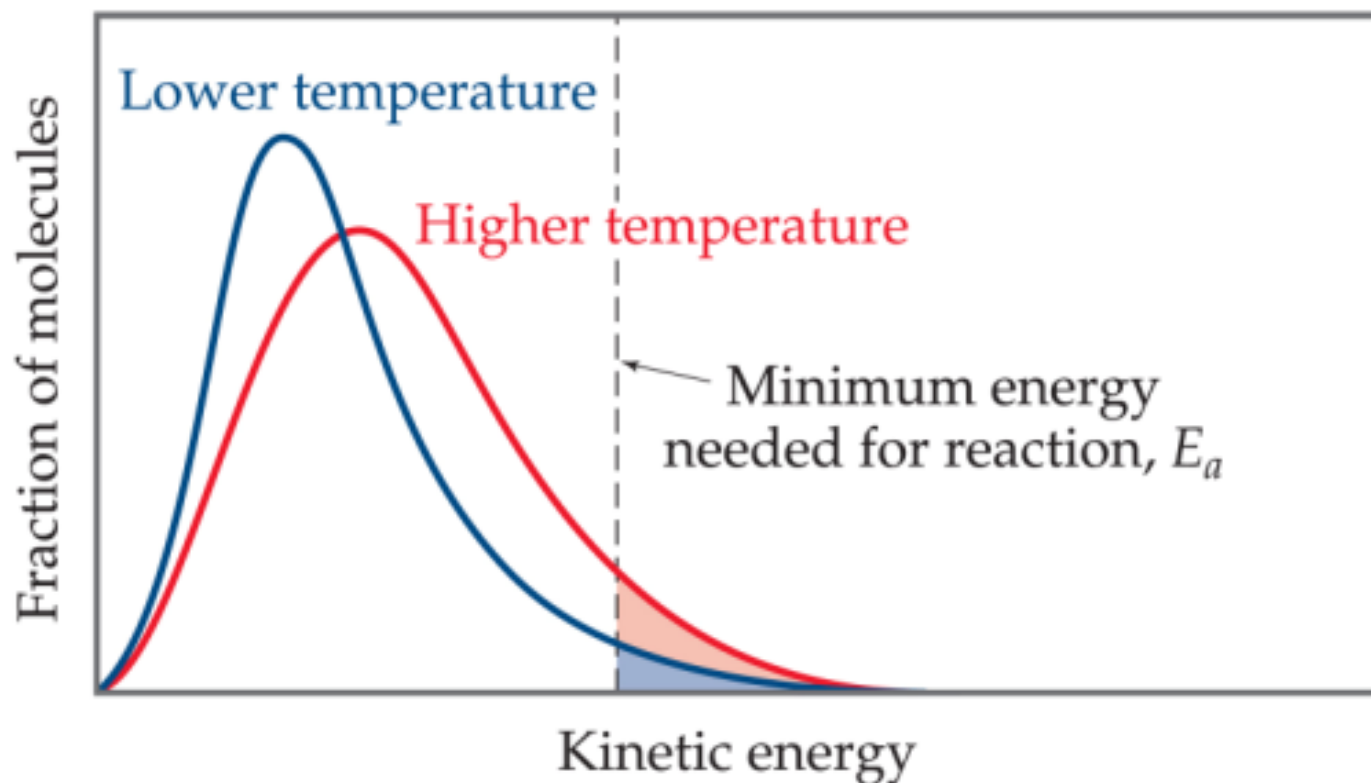
Orientation matters



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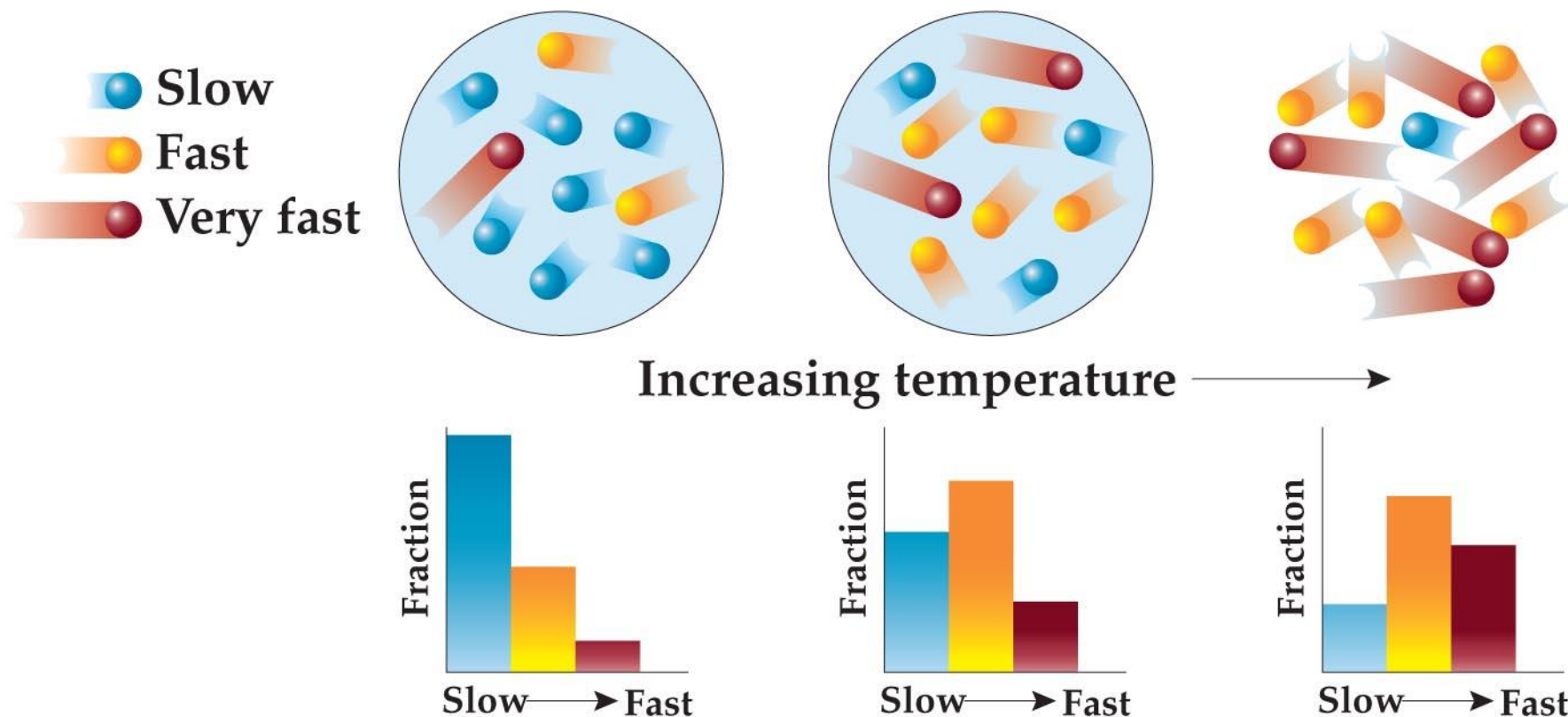
Higher Temperature, higher reaction rate



Rule of thumb: for every increase of 10°C , double reaction rate

Higher Temperature, higher reaction rate

The fraction of fast-moving molecules increases with increasing temperature.



Reaction rates

Any collision that is sufficiently energetic and properly oriented such that it leads to the formation of product is called an **effective collision**.

$$\text{Reaction rate} = \begin{array}{c} \text{Number of collisions} \\ \text{per unit time with} \\ \text{energy equal to or} \\ \text{greater than } E_a \end{array} \times \begin{array}{c} \text{Fraction of collisions in} \\ \text{which molecules have} \\ \text{proper orientation} \end{array}$$

Energy factor **Orientation factor**

Reaction rates

Reaction rate =

Molecules formed per second or Moles formed per second

Number of collisions per unit time with energy equal to or greater than E_a

Usually in collisions/s

X

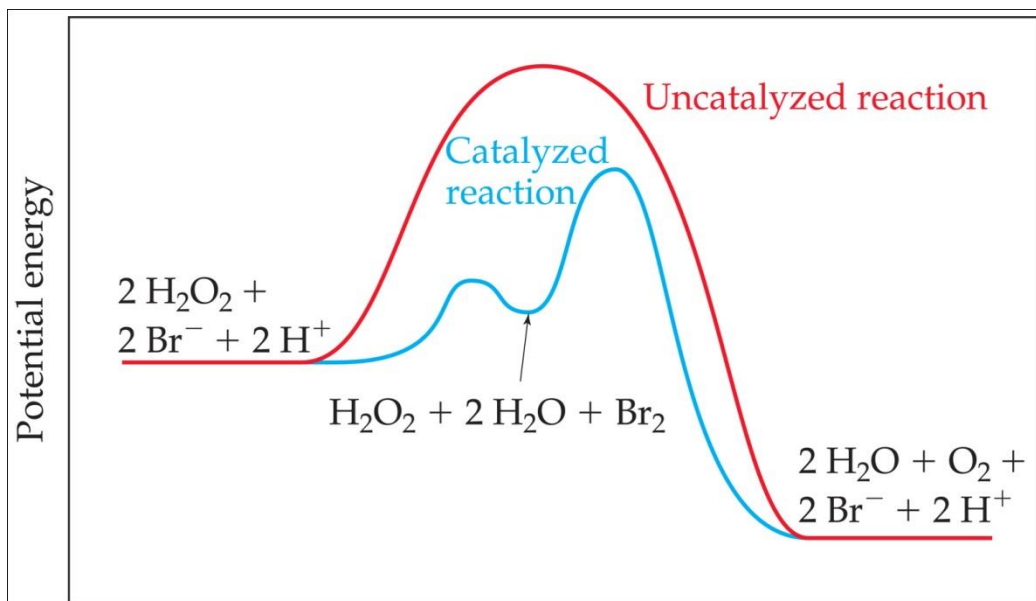
Fraction of collisions in which molecules have proper orientation

Orientation factor – between 0 and 1

$$\text{Reaction rate} = \left(\frac{\text{Molecules formed per second or Moles formed per second}}{\text{Number of collisions per unit time with energy equal to or greater than } E_a} \right) \times \left(\frac{\text{Fraction of collisions in which molecules have proper orientation}}{\text{Orientation factor – between 0 and 1}} \right)$$

Is there another way to make the reaction go faster?

A **catalyst** is a substance that changes (modifies) the rate of a reaction but does not undergo any permanent change.



Reaction pathway

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Usually changes both E_a and the orientation factor

The overall exothermicity and endothermicity of a chemical reaction is unaffected by the presence of a catalyst.

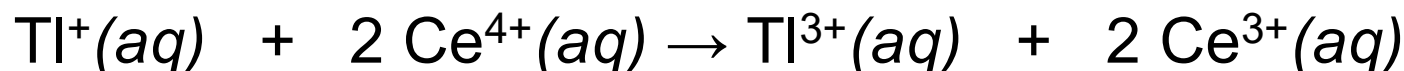
Catalyst example

student in dorm room → student in lecture hall

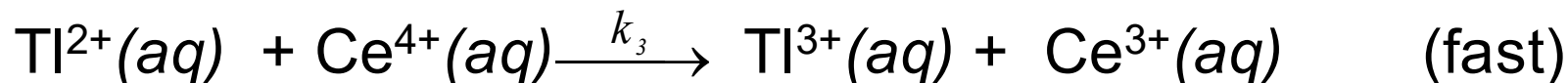
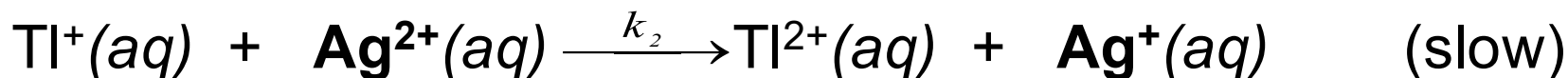
student in dorm room → student at 1st bus stop
student at 1st bus stop → student at 2nd bus stop
student at 2nd bus stop → student in lecture hall

student in dorm room → student at 1st bus stop
Bus + student at 1st bus stop → student at 2nd bus stop + Bus
student at 2nd bus stop → student in lecture hall

Catalyst example



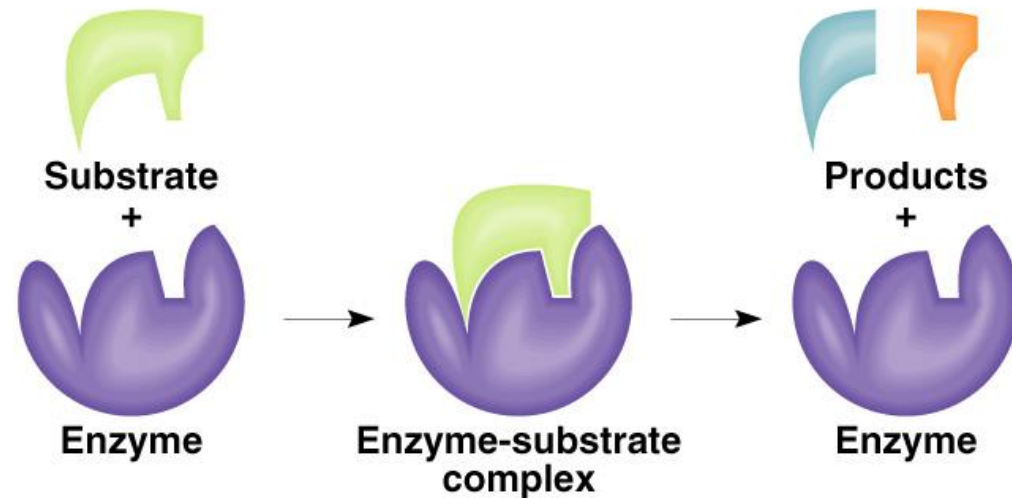
Catalyzed by Ag^+



Identifying catalyst and intermediates

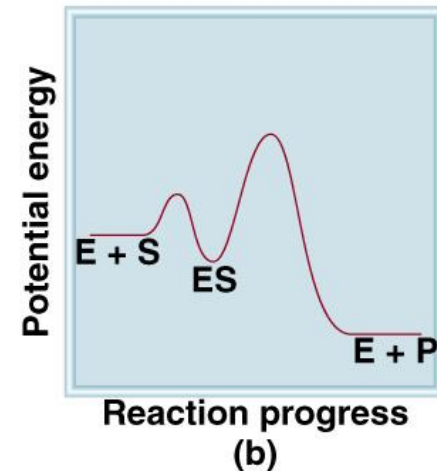
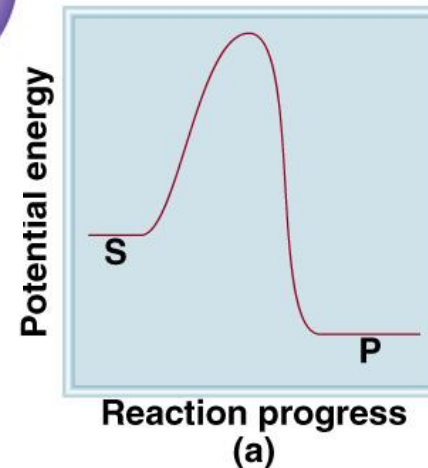
Another catalysis example - enzymes

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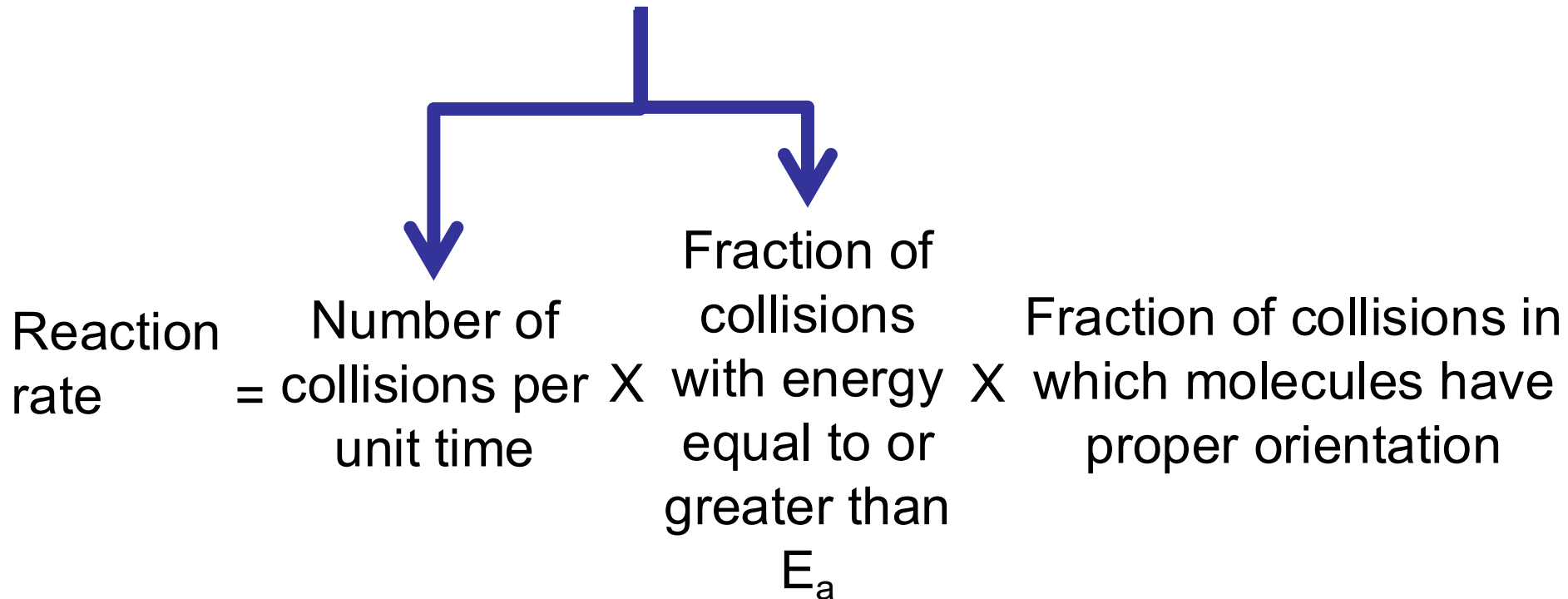
Lock and key mechanism

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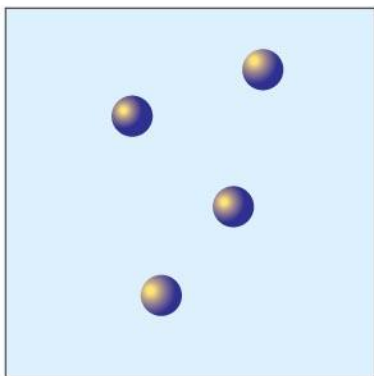
Reaction rates

$$\text{Reaction rate} = \text{Number of collisions per unit time with energy equal to or greater than } E_a \times \text{Fraction of collisions in which molecules have proper orientation}$$



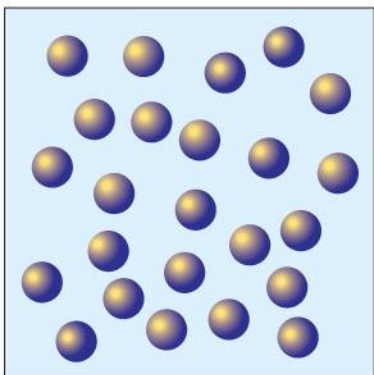
Both boxes are at the same temperature; the same reaction is occurring in both boxes; only 1% of collisions are effective.

Box 1



20 collisions/s

Box 2



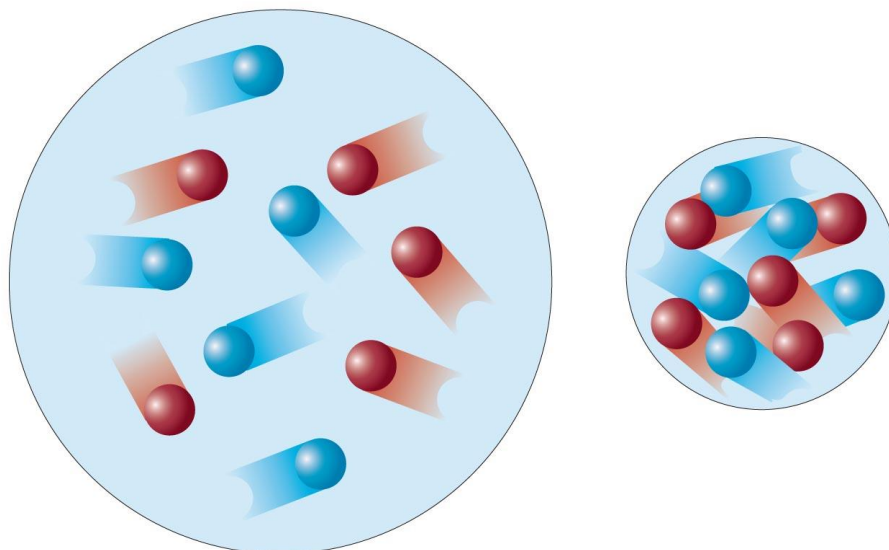
20,000 collisions/s

Dependence on concentration

The **greater** the concentration of **reactants**, the **greater** the total number of collisions per second and thus the **higher** the reaction rate.

Dependence on concentration

The number of collisions is proportional to the concentration – if the concentration increases, the number of collisions increase or if the concentration decreases, the number of collisions decrease



Reaction rates

Reaction rate = Number of collisions per unit time $\times$ Fraction of collisions with energy equal to or greater than E_a $\times$ Fraction of collisions in which molecules have proper orientation

Reaction rate = Number of collisions per unit time $\times$

k

Rate constant – independent of concentration

k is still dependent on temperature!!

Reaction Mechanisms

A reaction rarely involves only one step

A **reaction mechanism** is the detailed series of steps involved in a chemical reaction.

- Mechanisms are theoretical, cannot be directly observed, therefore cannot be proven. Although computational methods can now give important insight into these mechanisms
- Mechanisms are inferred from kinetics data
- Balanced reaction equations give the overall reaction or the net result of the reaction mechanism

Elementary reactions

An elementary reaction or elementary process refers to a single step in a reaction mechanism.

Represents the **actual** collision or decomposition happening

For an elementary reaction, the reactant orders in the rate law are determined by the stoichiometry of the balanced elementary reaction

Elementary reactions

Unimolecular elementary reaction: only 1 reactant (no collision)

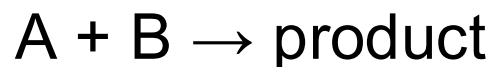


$$\text{rate} = k [A]$$

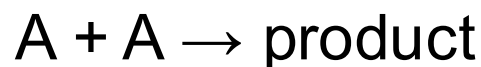
Remember: ONLY for an elementary reaction, the reactant orders in the rate law are determined by the stoichiometry of the balanced elementary reaction

Elementary reactions

Bimolecular elementary reaction: 2 molecules collide



$$\text{rate} = k [A][B]$$



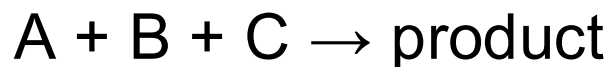
$$\text{rate} = k [A][A] = k [A]^2$$

Remember: ONLY for an elementary reaction, the reactant orders in the rate law are determined by the stoichiometry of the balanced elementary reaction

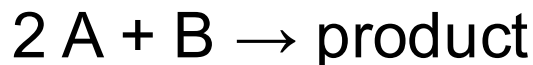
Elementary reactions

Termolecular elementary reaction: 3 molecules collide

RARE!



$$\text{rate} = k [A][B][C]$$



$$\text{rate} = k [A][A][B] = k [A]^2[B]$$

Remember: ONLY for an elementary reaction, the reactant orders in the rate law are determined by the stoichiometry of the balanced elementary reaction

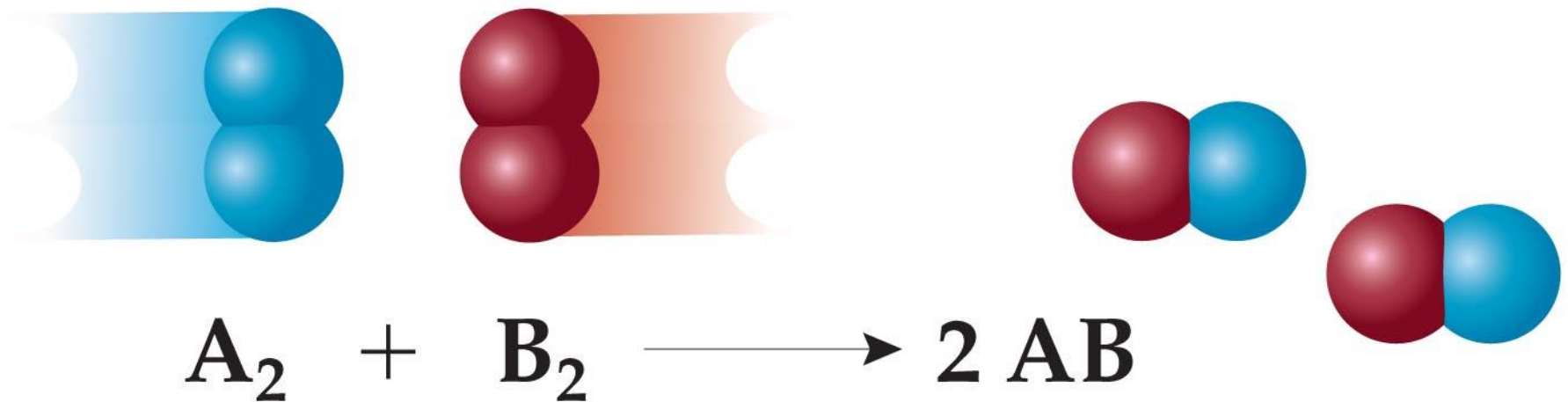
Multistep reactions

Consider the reaction $A_2 + B_2 \rightarrow 2 AB$

And suppose that kinetics experiments show the rate law to be

$$\text{Rate} = k [A_2]^2$$

Is the following elementary mechanism correct?



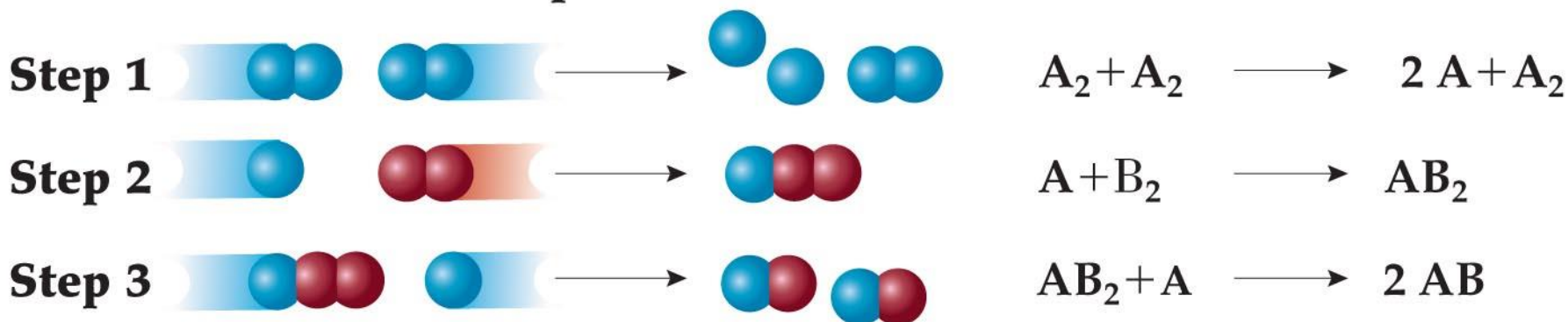
Multistep reactions

Consider the reaction $A_2 + B_2 \rightarrow 2 AB$

$$\text{Rate} = k [A_2]^2$$

Is the following multistep mechanism correct?

Mechanism II, three steps



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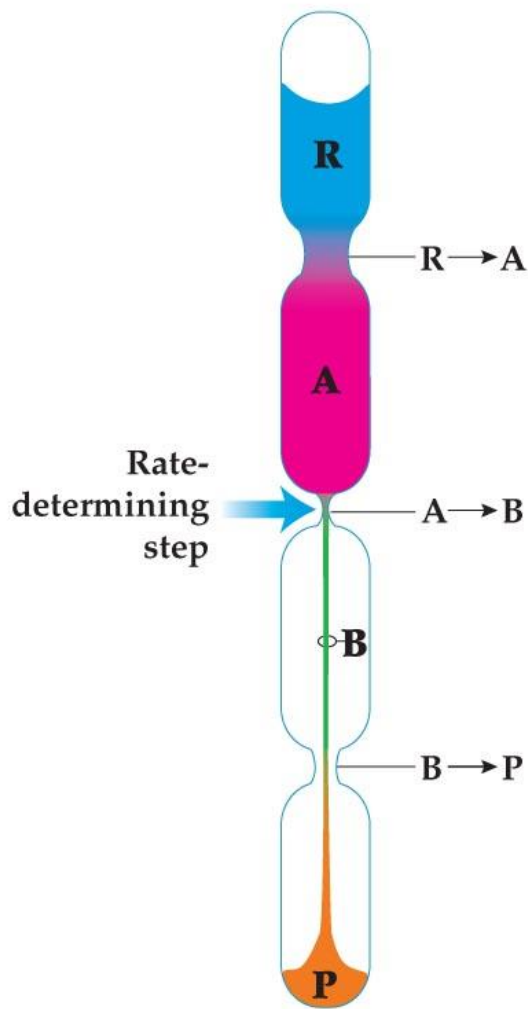
A and AB_2 are reaction intermediates

Maybe, if Step 1 is a rate limiting step

An Hourglass Illustrating the Concept of Rate-Determining Step

Overall reaction $R \longrightarrow P$

Mechanism $R \longrightarrow A \longrightarrow B \longrightarrow P$



The amount of P formed per second depends on how fast the rate-determining step ($A \longrightarrow B$) is.

Rate limiting or rate determining step

The **rate limiting step** is one step in a multistep reaction mechanism that is much slower than the others.

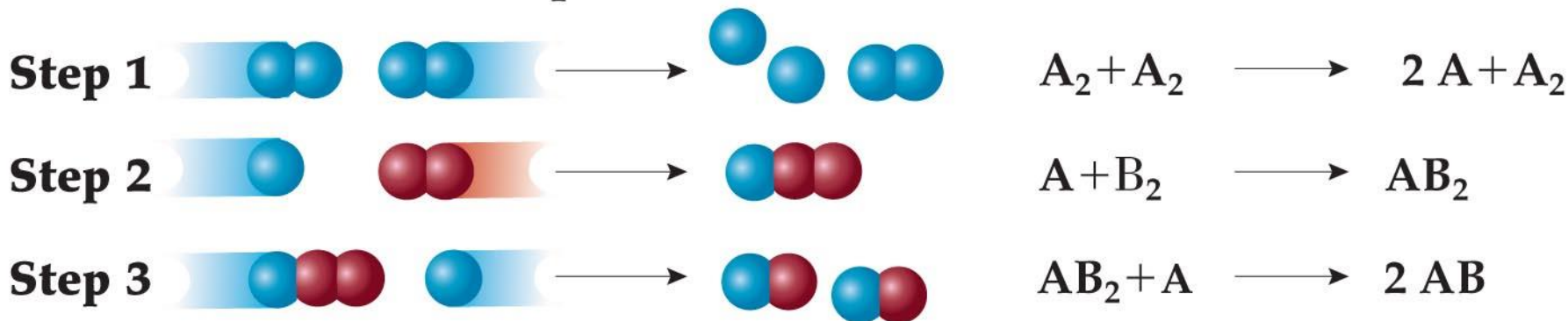
Multistep reactions

Consider the reaction $A_2 + B_2 \rightarrow 2 AB$

$$\text{Rate} = k [A_2]^2$$

Is the following multistep mechanism correct?

Mechanism II, three steps

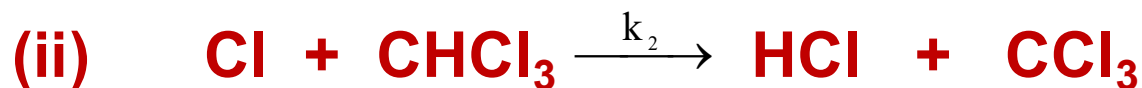
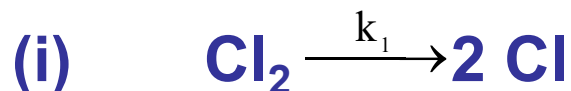


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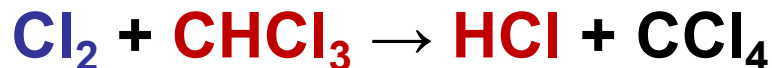
Maybe, if Step 1 is a **rate limiting step**

Another multistep mechanism

Consider the proposed mechanism

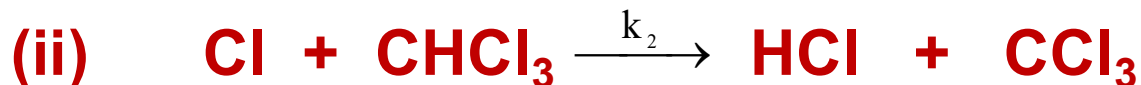
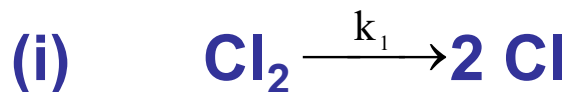


What is the overall reaction equation?



Another multistep mechanism

Consider the proposed mechanism



If the experimentally determined rate law is 1st order in Cl_2 and the concentration of CHCl_3 has no effect on the rate of the reaction, which step is the rate limiting step?

Summary of important rate law points

- Elementary steps should never have more than 2 species colliding simultaneously. Termolecular or 3-body collisions are extremely rare.
- The sum of the elementary steps must yield the overall balanced reaction equation for the mechanism to be valid.
- The mechanism must predict the experimentally observed rate law for it to be valid.
- If a mechanism is consistent with the experimental rate law, the mechanism is possibly correct. The mechanism is theoretical: it can't be proven but it can be disproven.

Extra materials

Balance the following reactions

1. When magnesium metal burns in air, it produces a white, powdery compound magnesium oxide.
2. When aluminum metal burns in air, it produces a white, powdery compound aluminum oxide.

Practice more examples

1. When ethyl alcohol ($\text{C}_2\text{H}_5\text{OH}$ – the liquor) is burnt in air, it produces carbon dioxide and water.
2. Combustion of liquid butane (C_4H_{10}) produces carbon dioxide and steam in a lighter.

Practice on end-of chapter problems: 8.26 through 8.42

Practice more examples

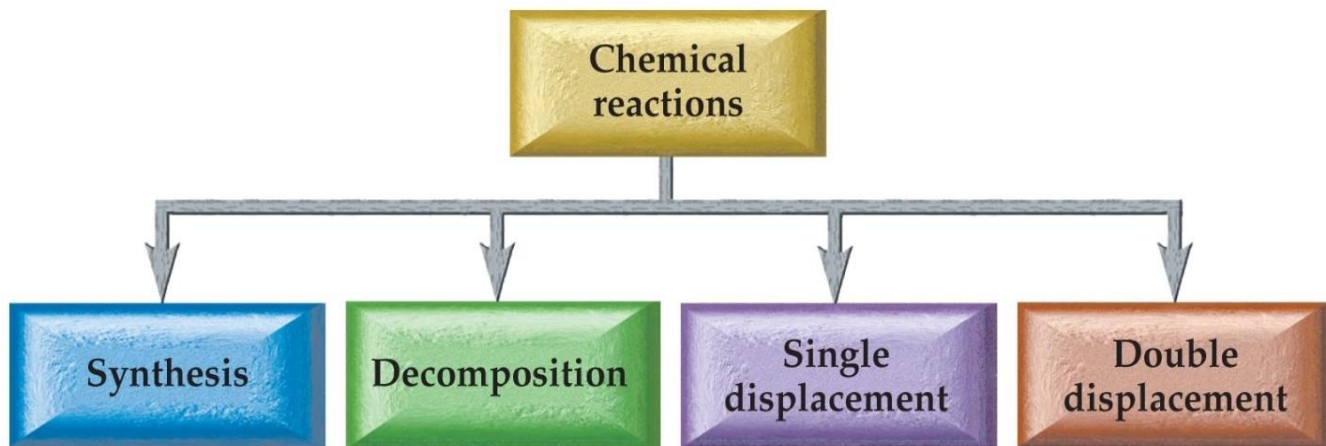
When methanol (CH_3OH) is burnt in air, it produces carbon dioxide and water.

Practice on end-of chapter problems: 8.26 through 8.42

Types of Reactions

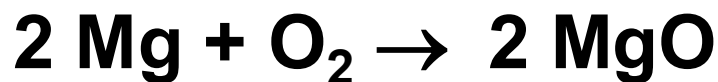
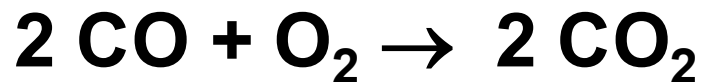
- One way of classifying the chemical reactions is by looking at what the atoms do.

Type of reaction	General equation	What atoms do?
Composition (synthesis)	$A + B \rightarrow AB$	Atoms combine
Decomposition	$AB \rightarrow A + B$	Atoms separate
Single Displacement	$A + BC \rightarrow AC + B$	Single atoms exchange
Double displacement	$AB + CD \rightarrow AD + CB$	Two atoms exchange



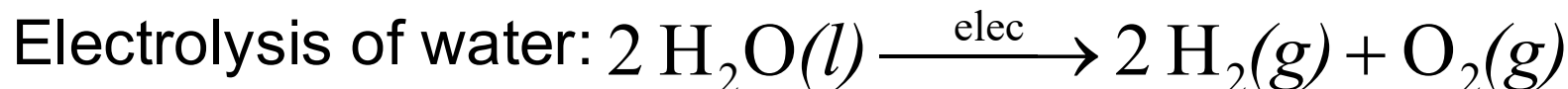
Combination Reactions

- Also known as **composition** or **synthesis** reactions.
- Two (or more) reactants combine together to make **one product**.
 - Simpler substances combining together.

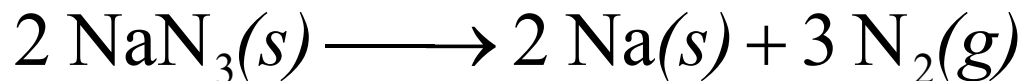


Decomposition Reactions

- A large molecule is broken apart into smaller molecules or its elements.
 - Caused by addition of energy into the molecule.



Function of Air-bags (decomposition of sodium azide, NaN_3):

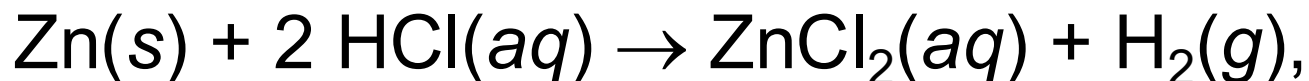


Note: the reverse of a combination reaction is a decomposition reaction and vice versa.

Single Displacement (or Replacement) Reactions

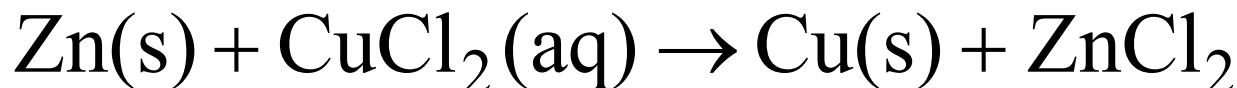
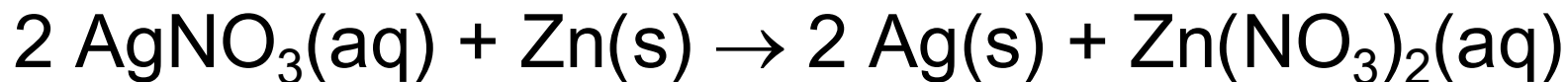
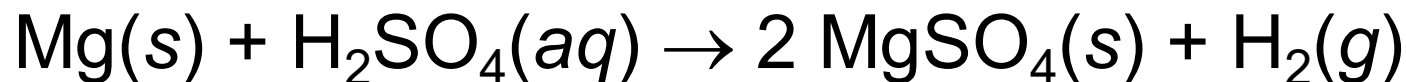
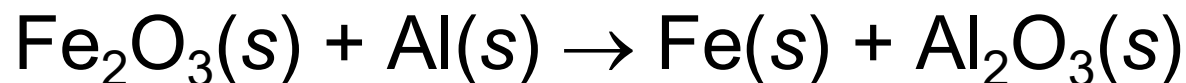
- Reactions that involve one atom displacing another and replacing it in a compound.

- In the reaction,

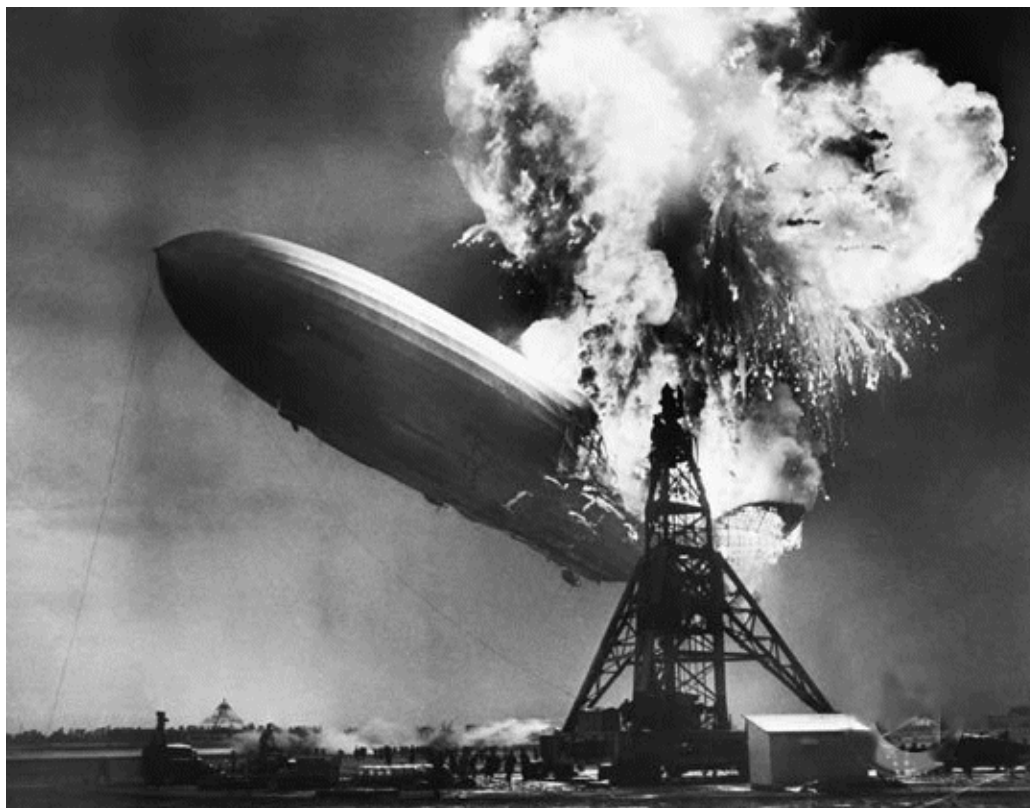
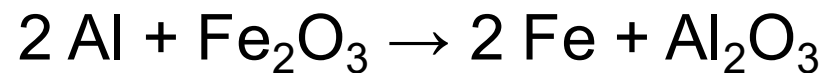


the atom Zn displaces H from the compound.

- Other examples of single displacement reactions are:



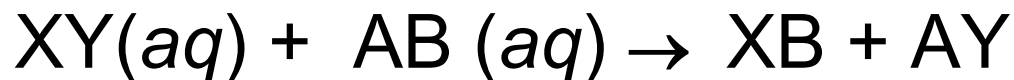
Thermite reaction



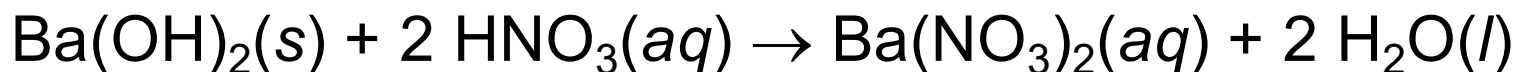
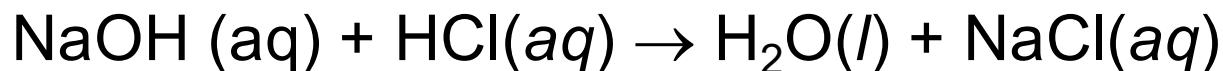
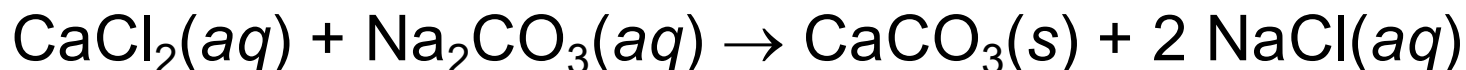
Hindenburg disaster May 6, 1937

Double Displacement (or Replacement) Reactions

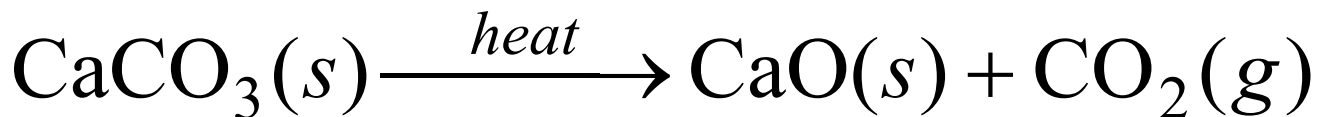
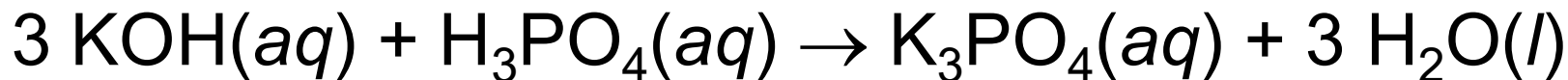
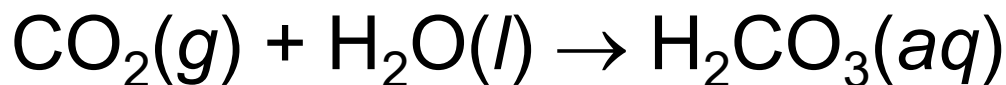
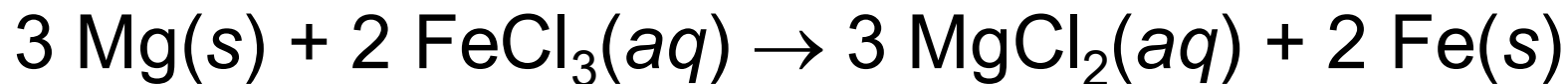
- Elements in two compounds exchange their places.



- Precipitation, acid–base, and gas evolving reactions are some examples of double displacement reactions.

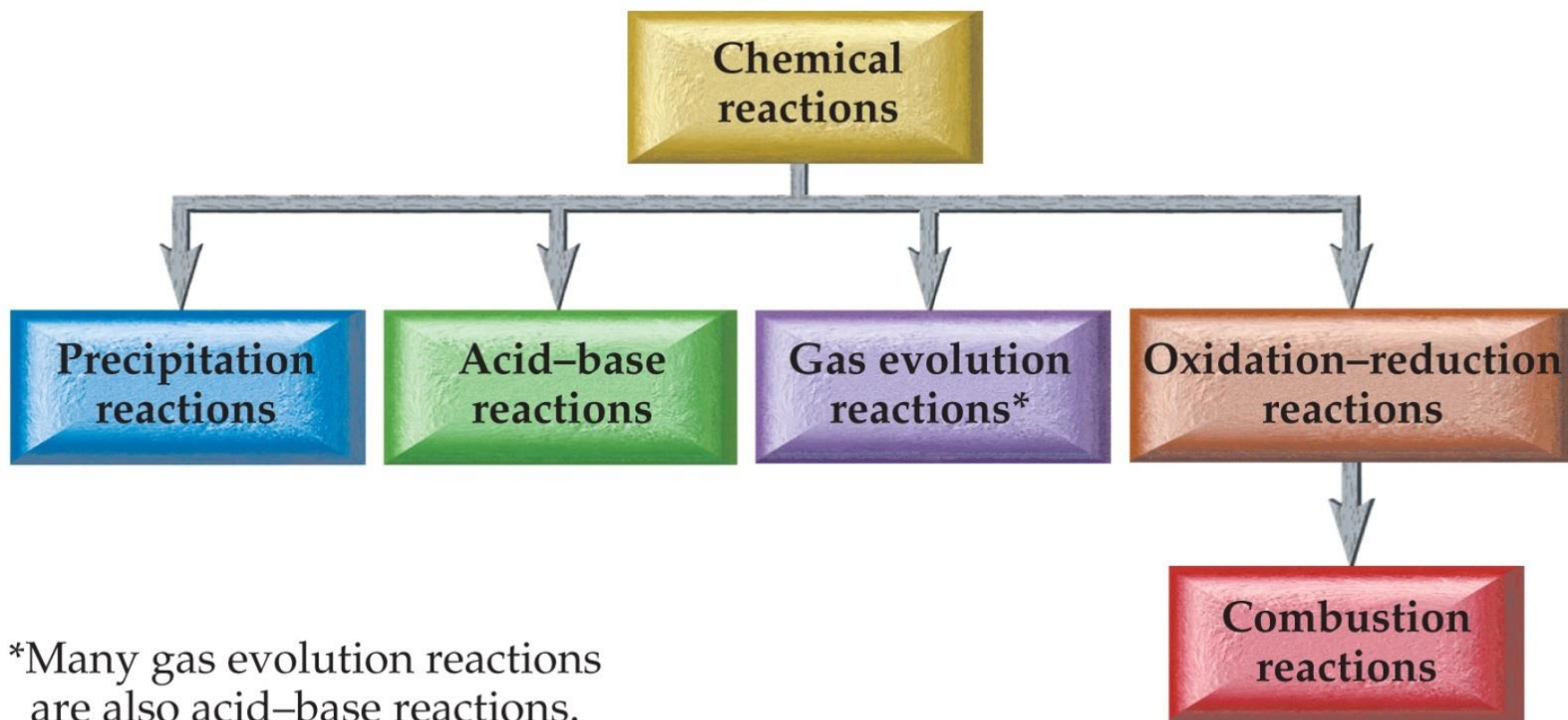


Practice—Classify the Following Reactions as Combination, Decomposition, Single Displacement, or Double Displacement.



Type of Reactions – Cont...

- Another way of classifying chemical reactions is based on the process that happens in a chemical reaction.
 - Precipitation, neutralization, formation of a gas, or transfer of electrons.



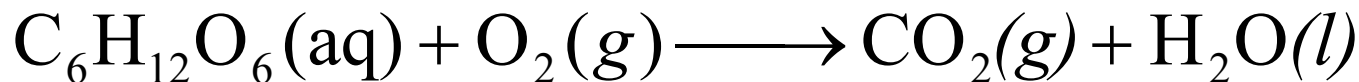
Combustion Reactions

- Reactions in which O_2 is consumed by combining with another substance are called **combustion reactions**.
 - Always release heat and/or other forms of energy.
 - Produce one or more oxygen-containing compounds.

Examples: see previous balancing equation slides

- In general, compounds containing only C, H and O in their composition produces CO_2 and H_2O as the only products in a combustion reaction.

Respiration (reaction of glucose in our body):



Try to balance this equation!

Solubility and Precipitation Reactions

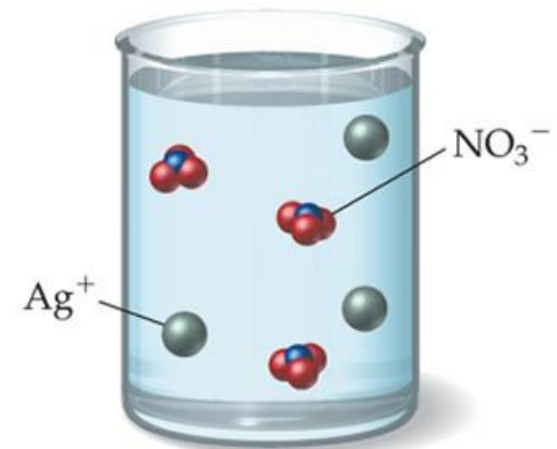
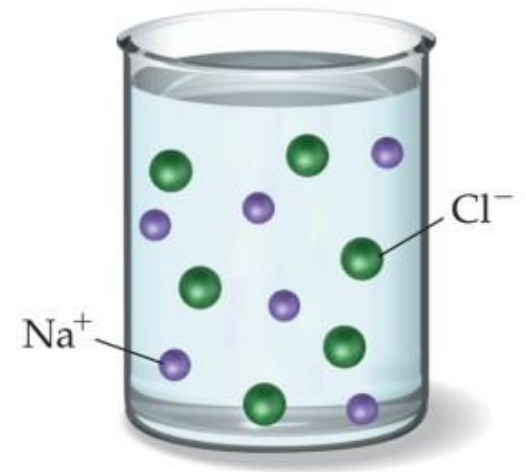
Reactions in Aqueous Solutions

- Many times, the chemicals we are reacting together are dissolved in water.
 - ✓ Mixtures of a chemical dissolved in water are called **aqueous solutions**.
- Dissolving the chemicals in water helps them to react faster.
 - ✓ The water (**aqueous medium**) separates the chemicals into individual molecules or ions.
 - ✓ The separate, free-floating particles come in contact more frequently so the reaction speeds up.

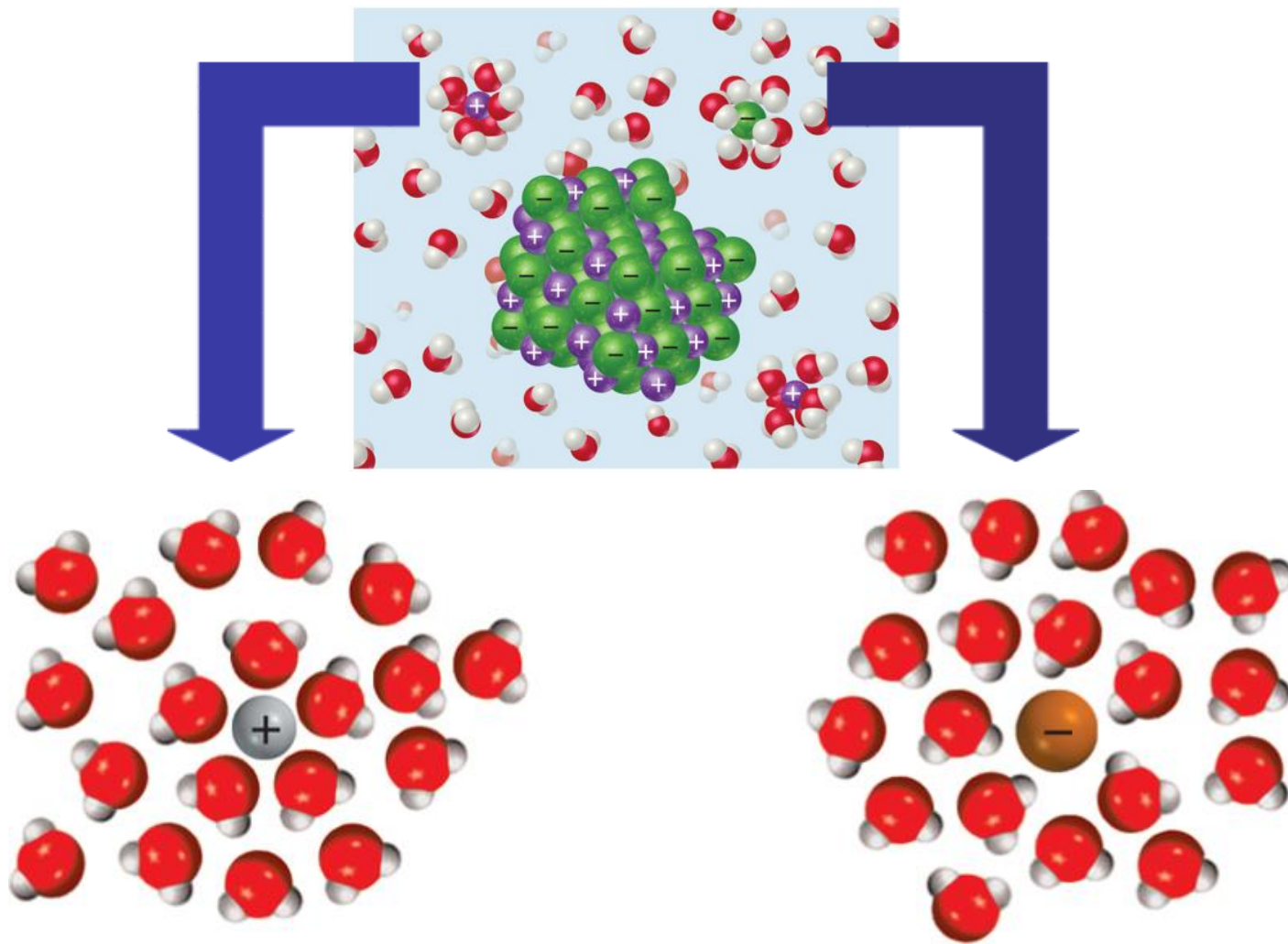
- When ionic compounds dissolve in water, the anions and cations are separated from each other. This is called **dissociation**.

✓ However, not all ionic compounds are soluble in water! (see following slides)

- When compounds containing **polyatomic ions dissociate**, the polyatomic group stays together as one ion.

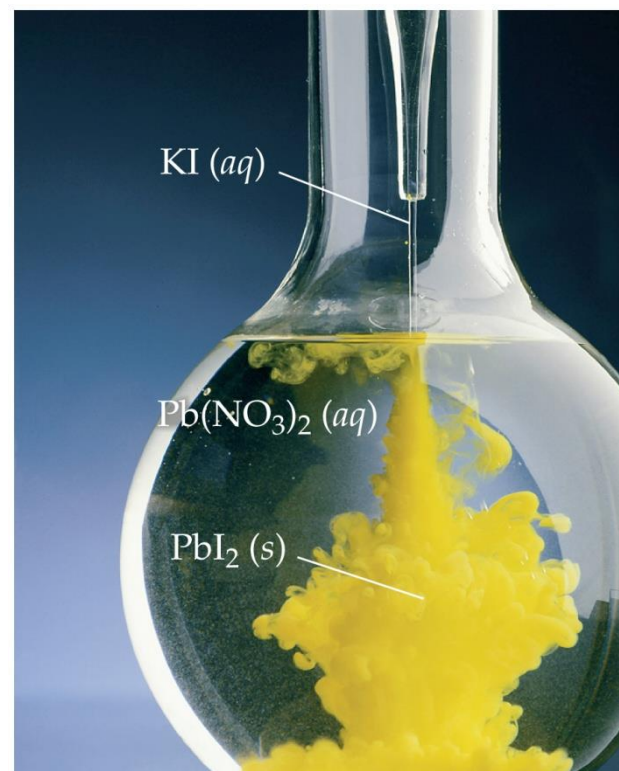


Why do ionic compounds dissociate in water?

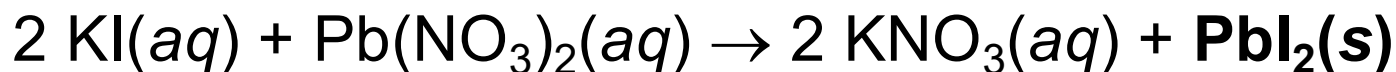


Precipitation Reactions

- Many precipitation reactions are done by mixing aqueous solutions of ionic compounds together.
- When this is done, often a reaction will take place from the cations and anions in the two solutions that are exchanging.
- If the ion exchange results in forming a compound that is insoluble in water, it will come out of solution as a **precipitate**.



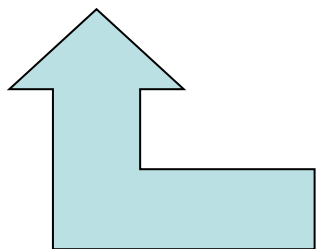
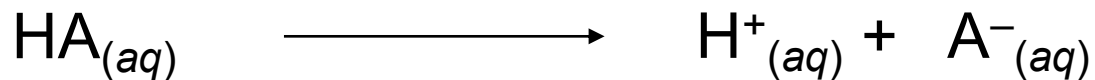
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(precipitate)

Acids and Bases

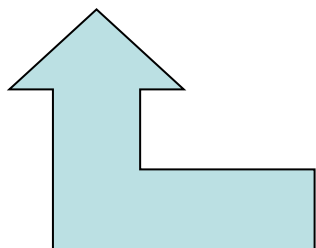
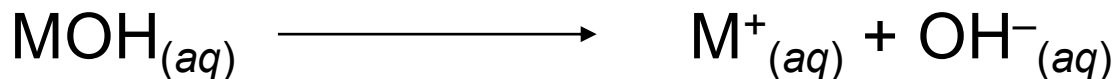
Acids in water give H^+



"general formula for acid"

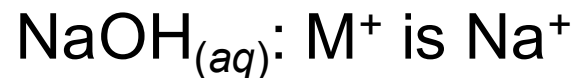


Bases in water give OH^-



"general formula for base"

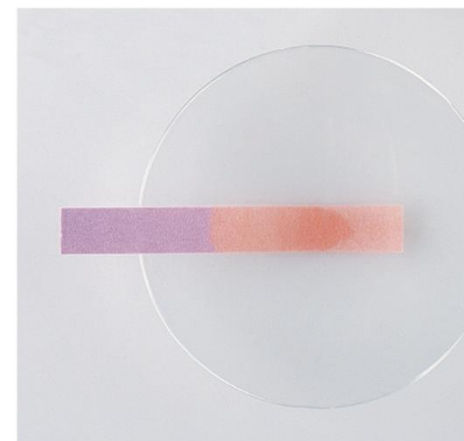
aqueous = dissolved
in water



Introduction to Acid-Base Reactions

Facts about Acids:

- Any substance that produce H^+ ions
 - Sour taste and changes the color of vegetable dyes.
 - React with “active” metals.
 - i.e., Al, Zn, Fe
- $$\text{Zn} + 2 \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$$
- Very corrosive.
 - React with carbonates, producing CO_2 .
 - Marble, baking soda, chalk, limestone.
- $$\text{CaCO}_3 + 2 \text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$$
- React with bases to form ionic salts.
 - And often water (neutralization)



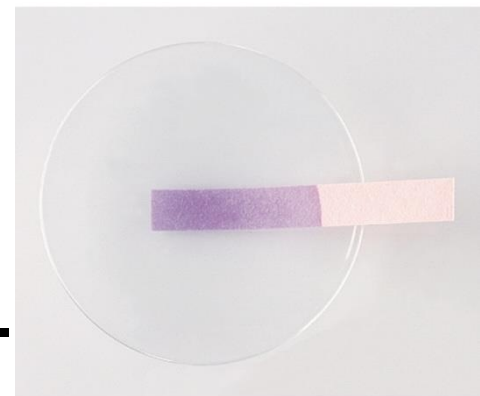
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Some Common Acids

Chemical name	Formula	Old name	Strength
Perchloric acid	HClO_4		Strong
Nitric acid	HNO_3	Aqua fortis	Strong
Sulfuric acid	H_2SO_4	Vitriolic acid	Strong
Hydrochloric acid	HCl	Muriatic acid	Strong
Phosphoric acid	H_3PO_4		Moderate
Chloric acid	HClO_3		Moderate
Acetic acid	$\text{HC}_2\text{H}_3\text{O}_2$	Vinegar	Weak
Hydrofluoric acid	HF		Weak
Carbonic acid	H_2CO_3	Soda water	Weak

Facts about Bases:

- Any substance that produce OH^- ions.
- A.k.a. alkalis.
- Taste bitter and feel slippery.
- Change color of vegetable dyes.
 - Different color than acid.
 - Litmus = blue.
- React with acids to form ionic salts.
 - And often water and the reaction is referred to as **Neutralization Reaction**.



Some Common Bases

Chemical name	Formula	Common name	Strength
Sodium hydroxide	NaOH	Lye, caustic soda	Strong
Potassium hydroxide	KOH	Caustic potash	Strong
Calcium hydroxide	Ca(OH) ₂	Slaked lime	Strong
Magnesium hydroxide	Mg(OH) ₂	Milk of magnesia	Weak
Ammonium hydroxide	NH ₄ OH, {NH ₃ (aq)}	Ammonia water, aqueous ammonia	Weak

Note that a lot of the cations are metals!!!